

N200 VIO Module User Manual V2.2.0

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Applicable to VIO software version V2.2.0

1. Description of Different Navigation Modes

To help users better understand and use the VIO module, the following section provides a brief explanation of different navigation modes.

1.1 VIO + GPS Navigation

During drone flight, positioning and navigation are fundamental to flight safety and mission execution. If your drone is equipped with both GNSS and VIO navigation, refer to the following explanations to understand how different flight controllers handle sensor fusion or switching.

1.1.1 PX4 Processing Mechanism

PX4 Autopilot adopts a multi-sensor fusion navigation strategy.

Under this mechanism, the flight controller can simultaneously receive data from multiple sensors, such as:

- GNSS positioning data
- Visual navigation (VIO) data
- Inertial sensor data (IMU)
- Barometer data

The flight controller fuses these inputs through internal algorithms to estimate the drone's current position and attitude.

Under this mechanism:

- When GNSS signals are reliable, the system uses satellite positioning as the global reference.
- When GNSS signals weaken or are lost, the system relies more heavily on VIO navigation.
- When both data sources are available, the system fuses both to improve positioning accuracy and stability.

This PX4 approach requires no manual intervention. The system automatically selects the most reliable data source, ensuring smooth and continuous navigation without noticeable switching.

1.1.2 APM Processing Mechanism

ArduPilot uses a different approach, primarily based on navigation source switching.

Multiple navigation sources can be preconfigured, such as:

- One GNSS navigation source
- One visual navigation (VIO) source

During operation, the system performs position estimation based on the navigation source currently selected by the user.

In the default APM configuration, when the flight environment changes, the APM flight controller will not automatically switch between GNSS and VIO. **Users need to manually switch the navigation source.**

1.2 GPS-denied VIO Navigation

When a drone is not equipped with a GPS module, additional configuration is required for the flight controller to establish a reference coordinate frame. Since visual positioning (VIO) provides relative position estimates referenced to the takeoff point, rather than global latitude and longitude coordinates, extra setup is necessary. Meanwhile, to ensure stable flight, the flight controller still relies on other sensors to provide essential state information.

1.2.1 Sensors Required for GPS-denied VIO Flight

Without GPS, the following sensors are still required:

Sensor	Function
Compass (Magnetometer)	Provides heading information
Barometer	Provides altitude information
IMU	Provides attitude information
N200M VIO	Provides relative position

Important:

The N200M module does not include a compass or barometer. Therefore, for GPS-denied flight, ensure these sensors are integrated into the flight controller hardware.

Without a compass or barometer, the flight controller cannot obtain complete flight state information, which may result in unstable flight.

1.2.2 PX4 Flight Controller Processing Mechanism

PX4 fuses VIO data through its EKF state estimation algorithm. When using PX4 without a GPS module installed, the drone's initial reference point must be manually set before flight (see Section 4.5).

Once configured, EKF continuously fuses these data inputs to estimate the drone's flight state for position control.

If visual data is temporarily lost, the system briefly relies on inertial navigation. Once visual data returns, EKF uses VIO to correct position errors.

1.2.3 ArduPilot Flight Controller Processing Mechanism

When using ArduPilot without GPS, the system also cannot automatically acquire global coordinates, so the flight origin must be manually set.

Users must also manually configure **ExternalNav (VIO)** as the primary position source to enable VIO navigation (see Section 5.2.2).

2. N200 Module Installation and Wiring

2.1 N200 Mounting Position

Install the module under the airframe as shown, ensuring:

- The downward-facing camera points vertically downward.
- The module is mounted as close as possible to the flight controller center.
- The arrow direction points toward the aircraft nose.

If the module cannot align with the nose direction, configure the **Camera Yaw** parameter in the web interface.

Use the configuration page to verify whether the camera view is obstructed. Any obstruction may reduce positioning accuracy.



2.2 N200 Interfaces

The module includes two interfaces:

1. UART Interface with power supply

Used for:

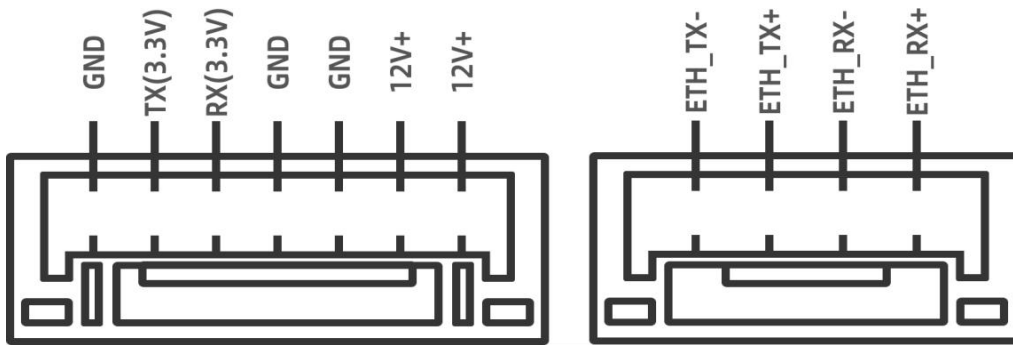
- powering the VIO module
- Communication with the flight controller .

2. Ethernet Interface

Used for:

- PC connection
- Module debugging
- Functional configuration

This interface is generally not used during normal operation.



2.3 Indicator LEDs



Indicator Light

Solid On:

Flight controller connection is normal, and the module is operating properly.

Fast Flashing:

Flight controller communication abnormal.

Slow Flashing:

Module fault.

Off:

No power.

3. Introduction of VIO Configuration Web Interface

Users can use the VIO web interface to:

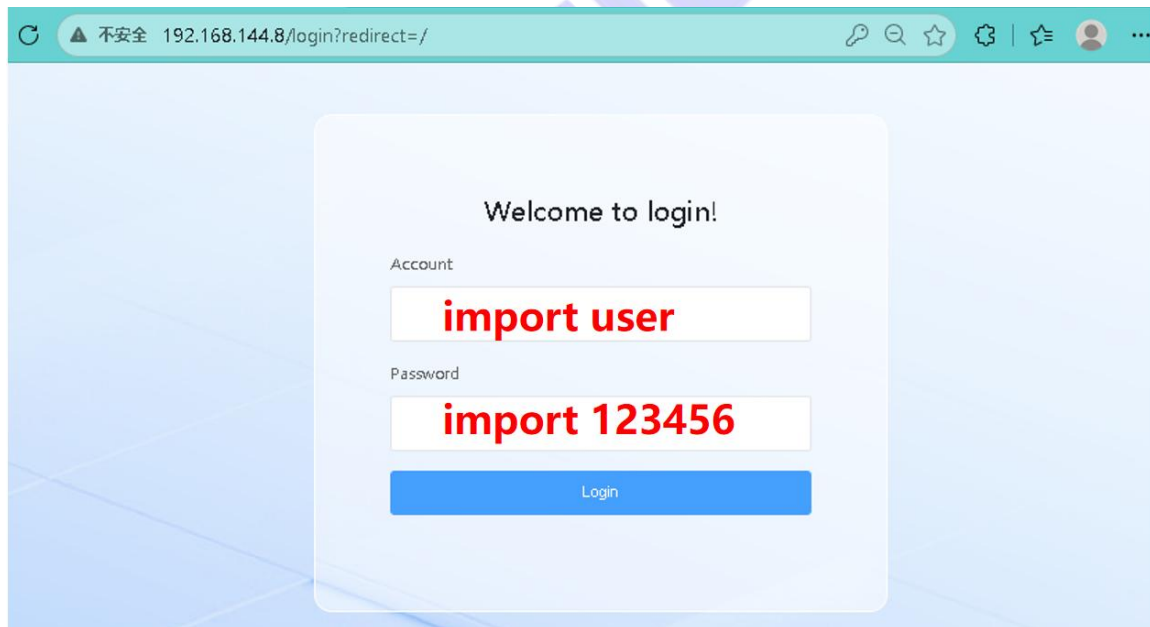
- View VIO module information
- Verify module status
- Configure flight controller settings
- Export logs

3.1 Login

1. Ensure VIO is powered properly through the UART power line. Connect the supplied Ethernet cable between the module and PC.
2. Set the PC IP to another address in the same subnet, for example: 192.168.144.2
3. Then open: <http://192.168.144.8>

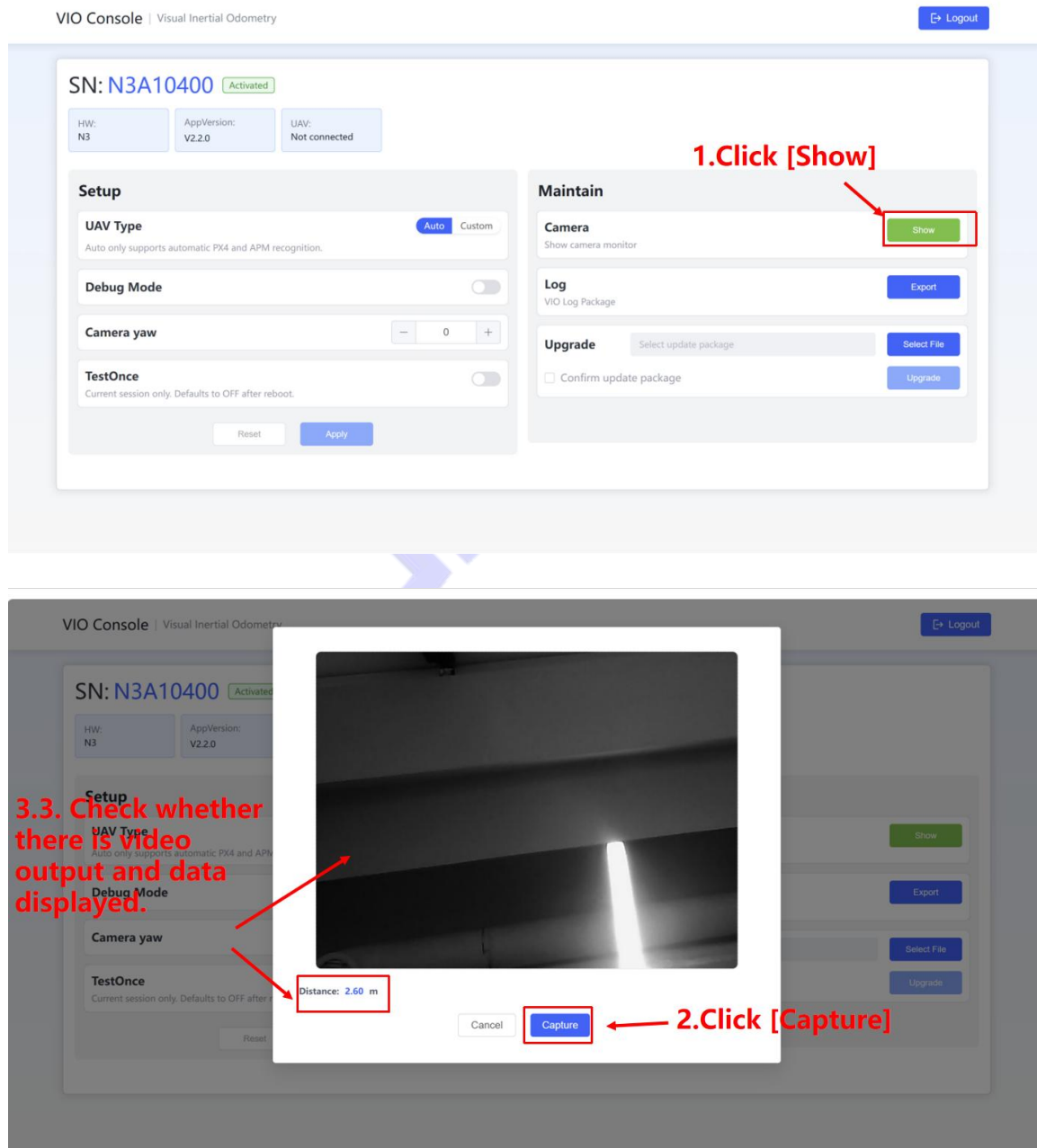
Login credentials:

- Username: user
- Password: 123456

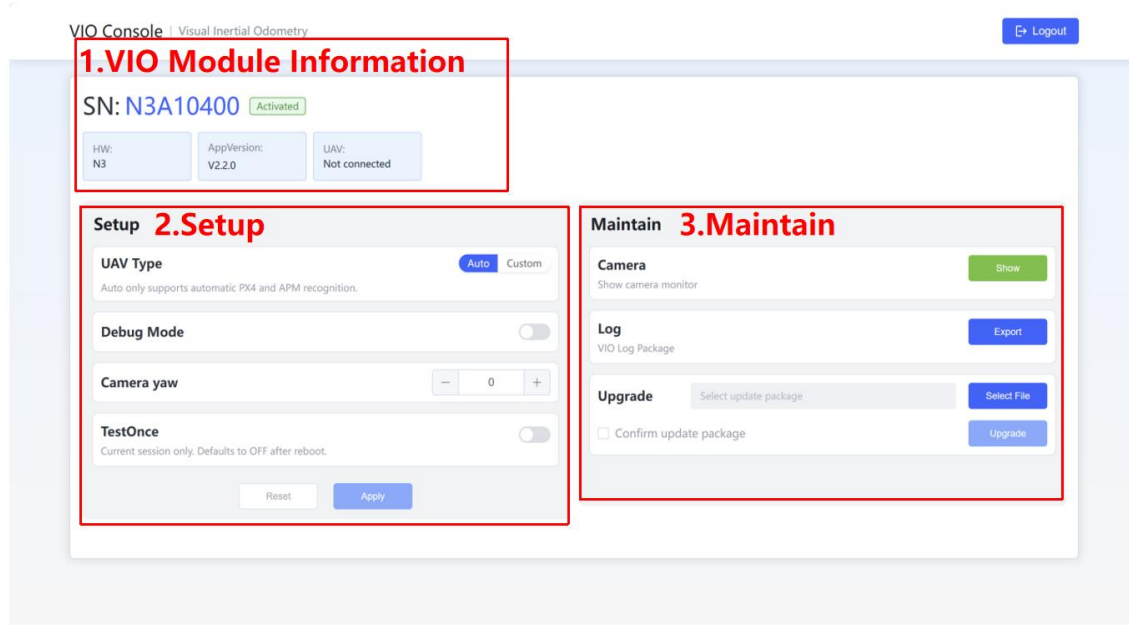


3.2 Status and Occlusion Check of VIO

1. After receiving the VIO module, you can verify whether the module is functioning properly through the image feed; click **【Show】** to open the display window, then click **【Capture】** to take a photo and check the laser rangefinder reading.
2. When confirming the VIO module installation position, use the image to check whether the module's view is obstructed. If the image is blocked, change the installation position or remove the obstruction, otherwise positioning accuracy may be affected.



3.3 Page Description



1. VIO Module Information

- **SN:** Factory serial number (matches label on device).
- **HW Version:** Hardware version.
- **APP Version:** Software version.
- **UAV:** Displays the flight controller type. The V2.2.0 VIO module supports automatic detection of both PX4 and APM
- **UAV Version:** Flight controller firmware version read after successful connection between the drone and the VIO module

2. Setup

- **UAV Type:** Flight controller type.
The default setting is Auto. After connecting to the flight controller, the VIO module will automatically detect the controller type and configure certain PX4 or APM parameters accordingly.
- **Debug Mode:** Disabled by default. Users do not need to configure this parameter.
- **Camera Yaw:** Camera offset angle. The default value is 0°.
If there is an angle offset between the VIO module orientation and the drone heading, this parameter must be configured.

Angle definition:

Keep the drone nose facing forward and view from top to bottom:

- If the module is rotated to the left relative to the drone heading, **camera_yaw** is negative.
- If the module is rotated to the right relative to the drone heading, **camera_yaw** is positive. The value should be the actual offset angle in degrees (°).

After modification, click **Apply** to save the parameters, **then restart the VIO module**.

- **TestOnce:**

To simplify integration testing, the configuration page provides a **TestOnce** function.

- i. **Disabled:**

Default state. In this mode, the VIO module outputs visual positioning data only when the altitude is above 3 meters. Each time the VIO module restarts, TestOnce automatically returns to the disabled state to avoid positioning drift caused by unstable visual data during low-altitude flight.

- ii. **Enabled:**

Removes the altitude restriction for positioning output, allowing visual positioning data to be output even below 3 meters.

Note: Visual performance may become unstable below 3 meters, which can result in positioning drift. Flight testing below 3 meters is not recommended. The TestOnce function is valid only for the current session and will return to the default disabled state after reboot.

3. Maintain

- **Camera:** View the camera image and laser distance readings.
- **Log:** When exporting VIO module logs is required, click the **Export** button, select the corresponding date, and download the VIO module logs.
- **Upgrade:** Used to upgrade the VIO module firmware version. Click the **Select File** button, choose the installation package from the local directory, check **Confirm update package**, then click the **Upgrade** button.

4. Connecting VIO to PX4 Flight Controllers

This module currently supports PX4 version V1.15 and above. Configuring the connection parameters between the PX4 flight controller and the VIO module involves logging into the VIO web interface and using the QGroundControl software.

Users should first ensure that the VIO module is properly powered, the communication serial port is connected to the flight controller, the VIO web interface has been successfully accessed, and the flight controller is connected to QGC.

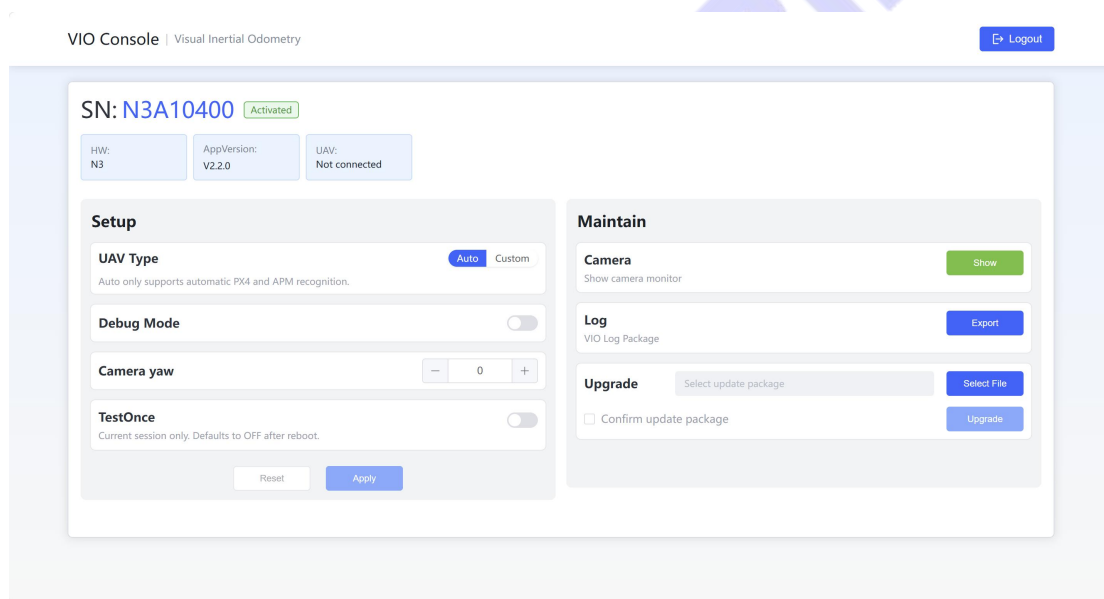
4.1 VIO Parameter Configuration

After logging into the VIO configuration webpage:

Camera Yaw: Enter the actual angle offset between the VIO module and the aircraft nose. If there is no offset, no modification is required.

Other parameters: Keep the default settings; no changes are needed.

If any parameters are modified, click **Apply** to save them, **then power off and restart the VIO module. The updated settings will only take effect after the module reboots.**



The screenshot displays the VIO Console web interface. At the top, it shows 'VIO Console | Visual Inertial Odometry' and a 'Logout' button. Below this, the system status is shown: SN: N3A10400 (Activated), HW: N3, AppVersion: V2.2.0, and UAV: Not connected. The interface is divided into two main sections: 'Setup' and 'Maintain'. The 'Setup' section includes 'UAV Type' (Auto/Custom), 'Debug Mode' (toggle), 'Camera yaw' (input field with - 0 +), and 'TestOnce' (toggle). The 'Maintain' section includes 'Camera' (Show button), 'Log' (Export button), and 'Upgrade' (Select update package, Select File, and Upgrade buttons). At the bottom of the Setup section are 'Reset' and 'Apply' buttons.

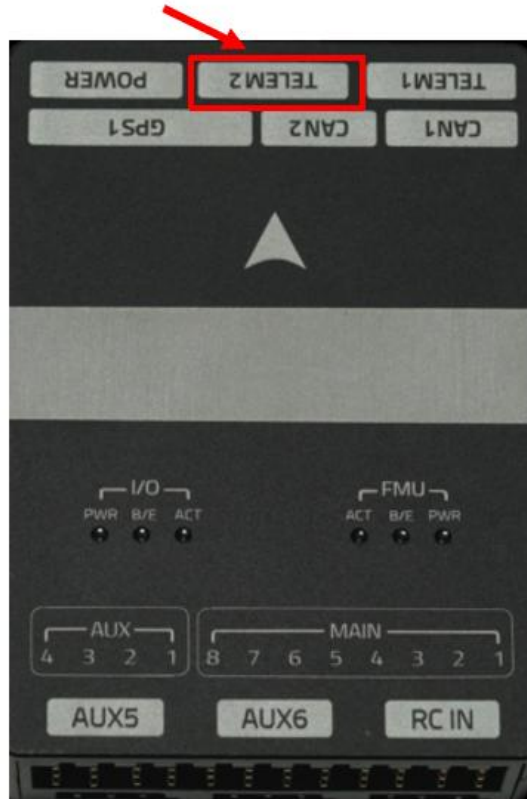
4.2 Flight Controller Communication Parameters

4.2.1 Confirm PX4 Serial Port

Because different PX4 flight controllers (such as **Pixhawk 4**, **Orange Cube**, and **Pixhawk 6C**) may use different physical port labels and internal port identifiers, users should first determine which serial port on their flight controller is connected to the VIO module.

For ease of explanation, the following example assumes the VIO module is connected to the flight controller's **TELEM 2** port to illustrate how to configure the connection between PX4 and VIO.

Assume the VIO connects to TELEM 2

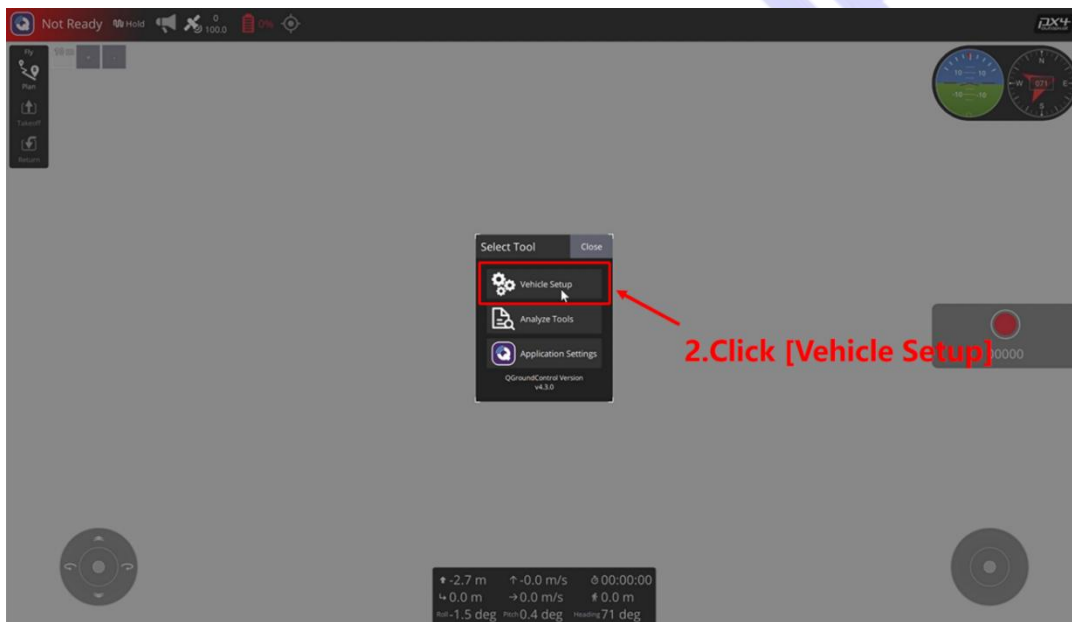
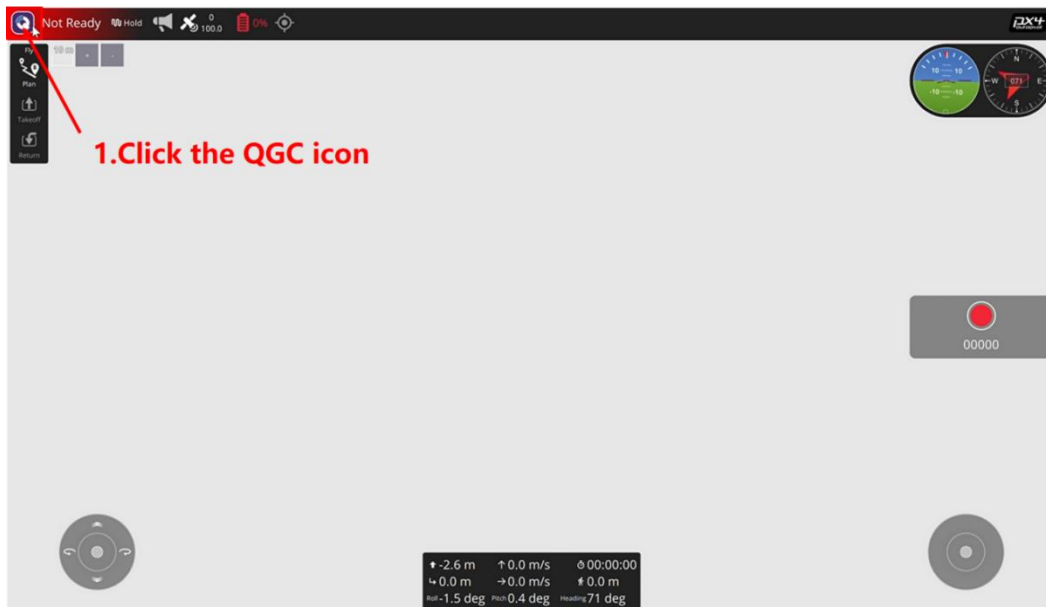


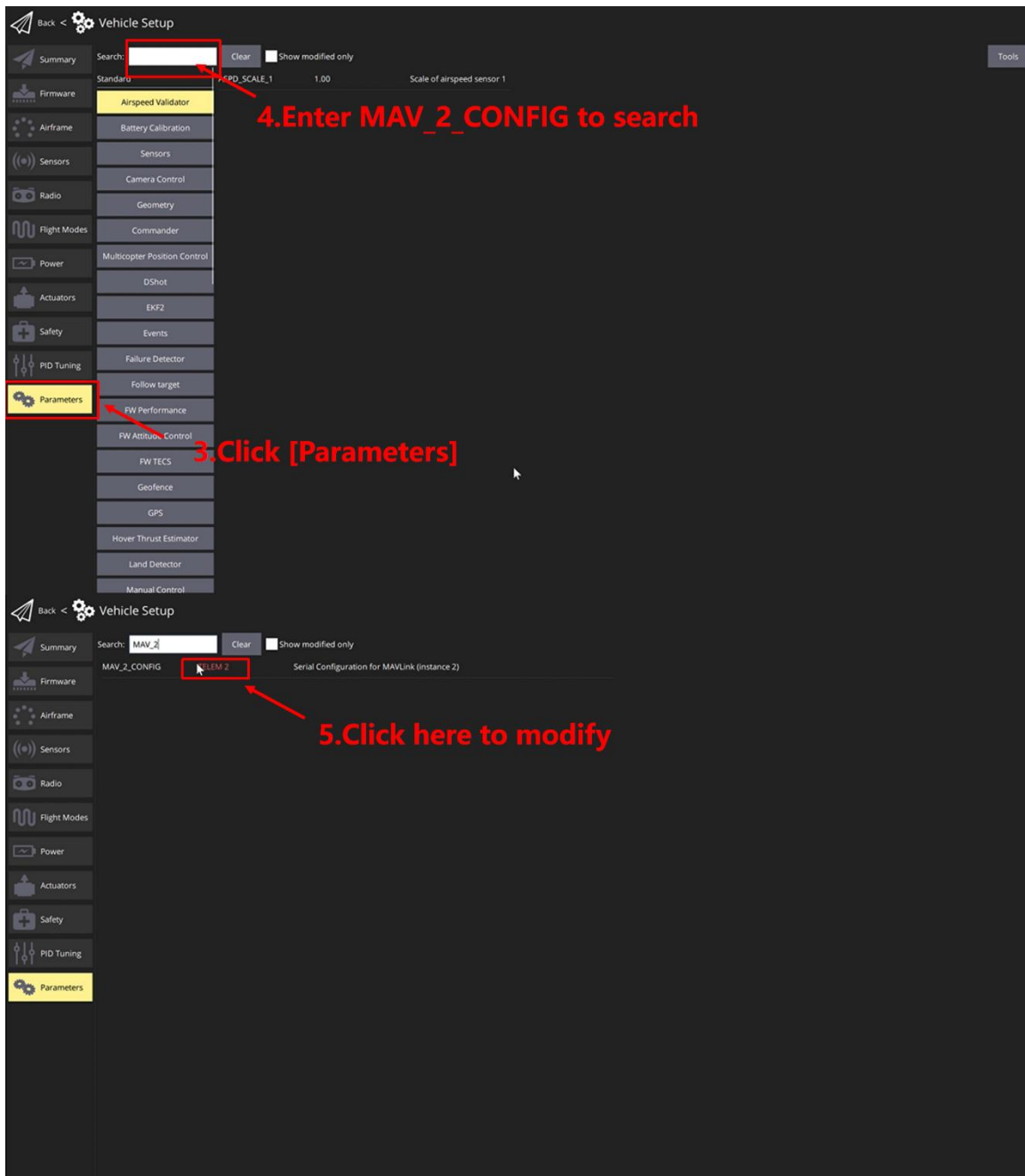
4.2.2 Configure MAVLink Instance

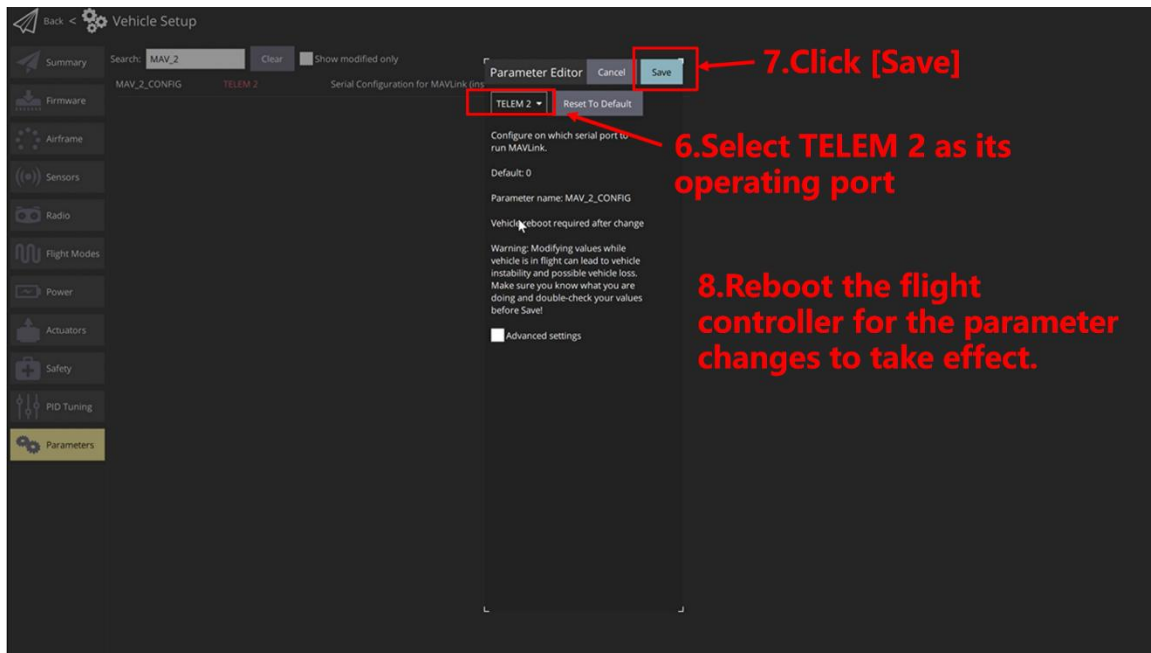
First, a MAVLink instance must be made to transmit visual positioning data. Assuming MAVLink instance 2 is available, set **MAV_2_CONFIG** to **TELEM 2**.

In **QGC** → **Vehicle Setup** → **Parameters**, search for **MAV_2_CONFIG** and select **TELEM 2** as its operating port.

After completing the configuration, **reboot the flight controller for the parameter changes to take effect.**



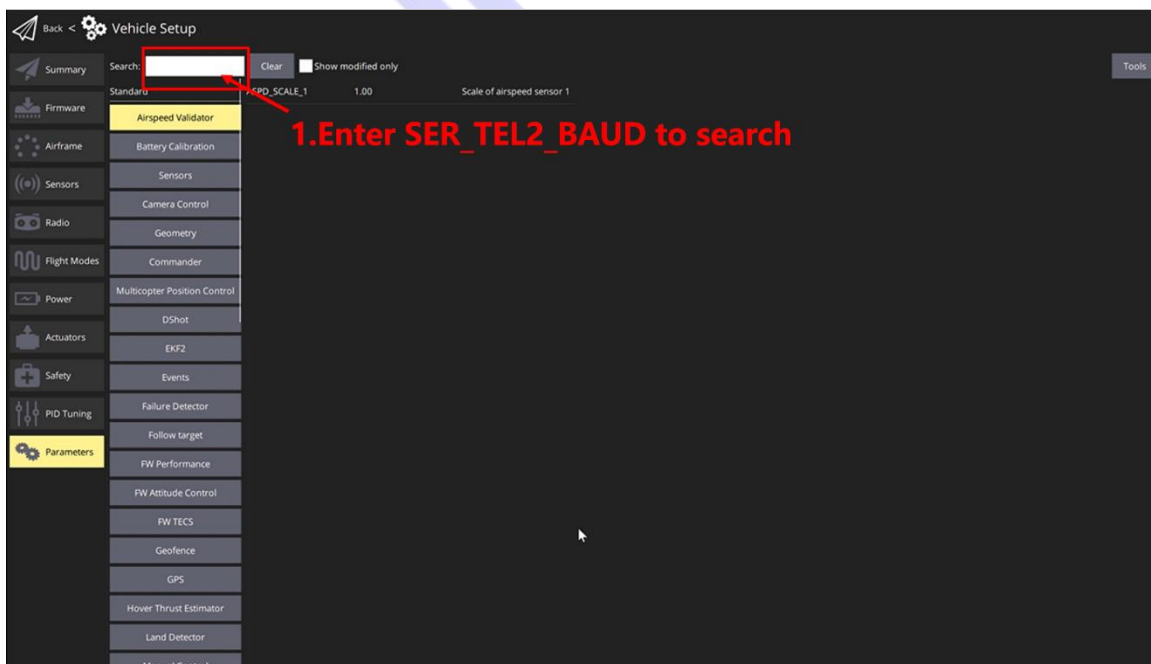


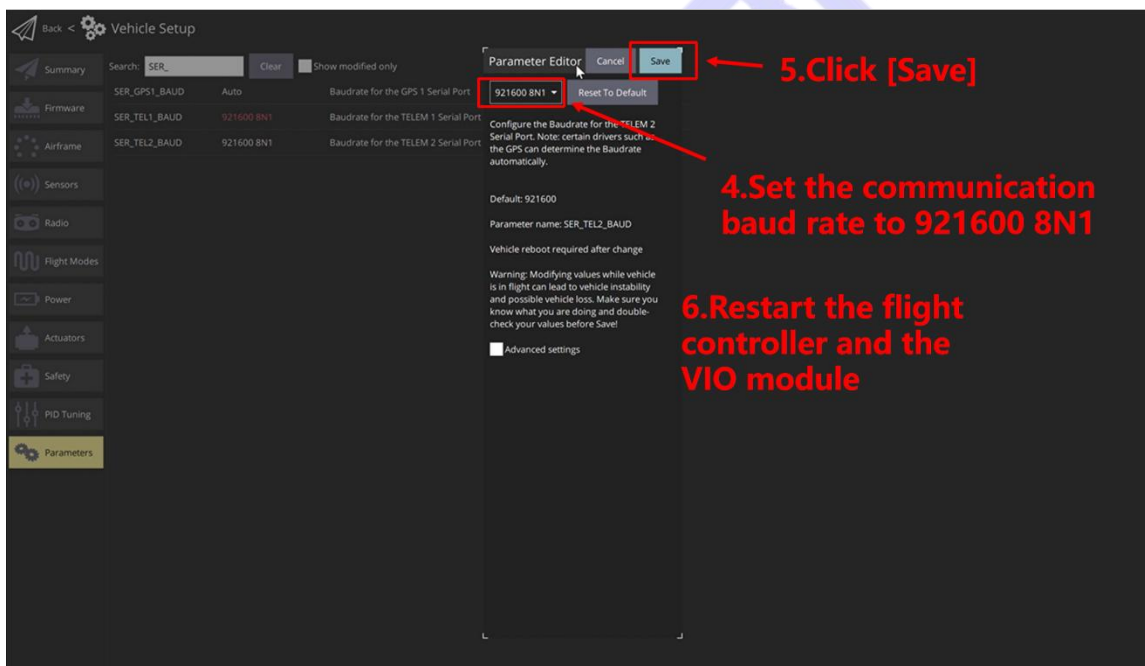
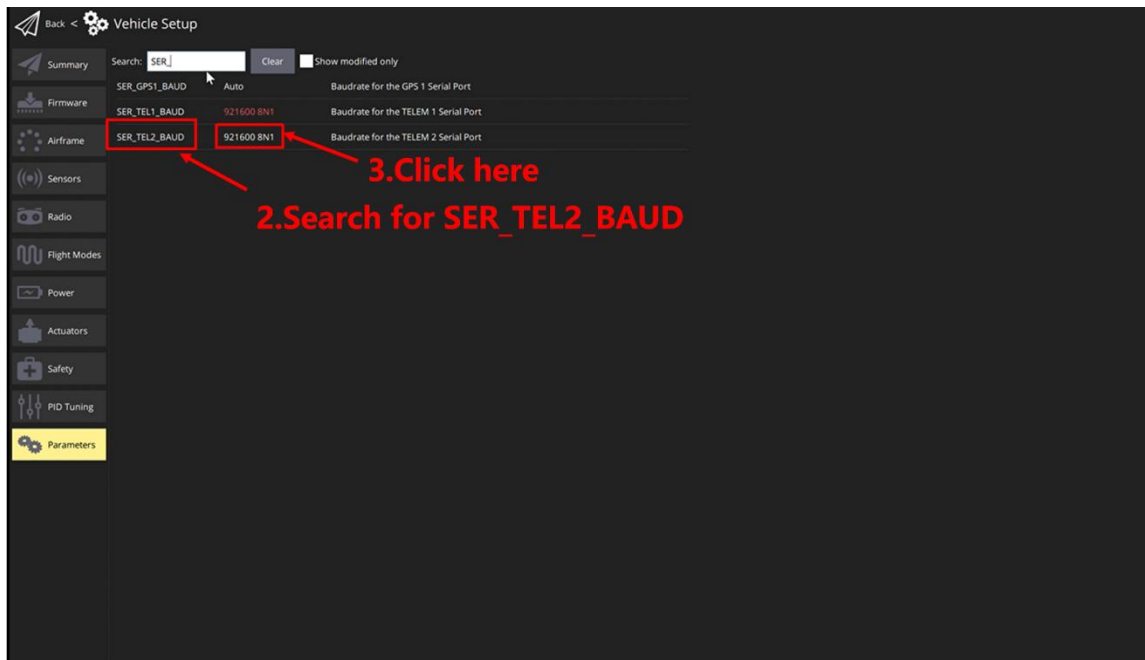


4.2.3 Set Baud Rate

In QGC → Vehicle Setup → Parameters, search for **SER_TEL2_BAUD** and set the communication baud rate to **921600 8N1**.

After completing the configuration, reboot both the flight controller and the VIO module.





4.3 Connection Status Verification

4.3.1 VIO Side Diagnostics

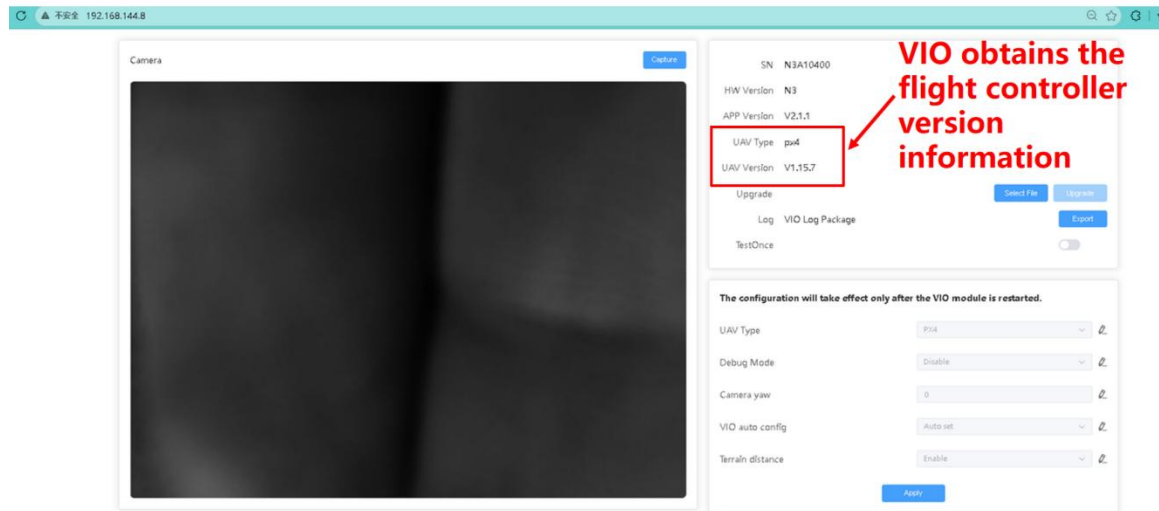
Verify the VIO web interface can read the following information (as show on the picture below):

- PX4 firmware version

- Flight controller information

If successful, the VIO is successfully connected to the flight controller and communication is normal.

If no flight controller information is received, recheck the serial wiring order, baud rate, and whether the standard MAVLink protocol is enabled on the flight controller as described above.



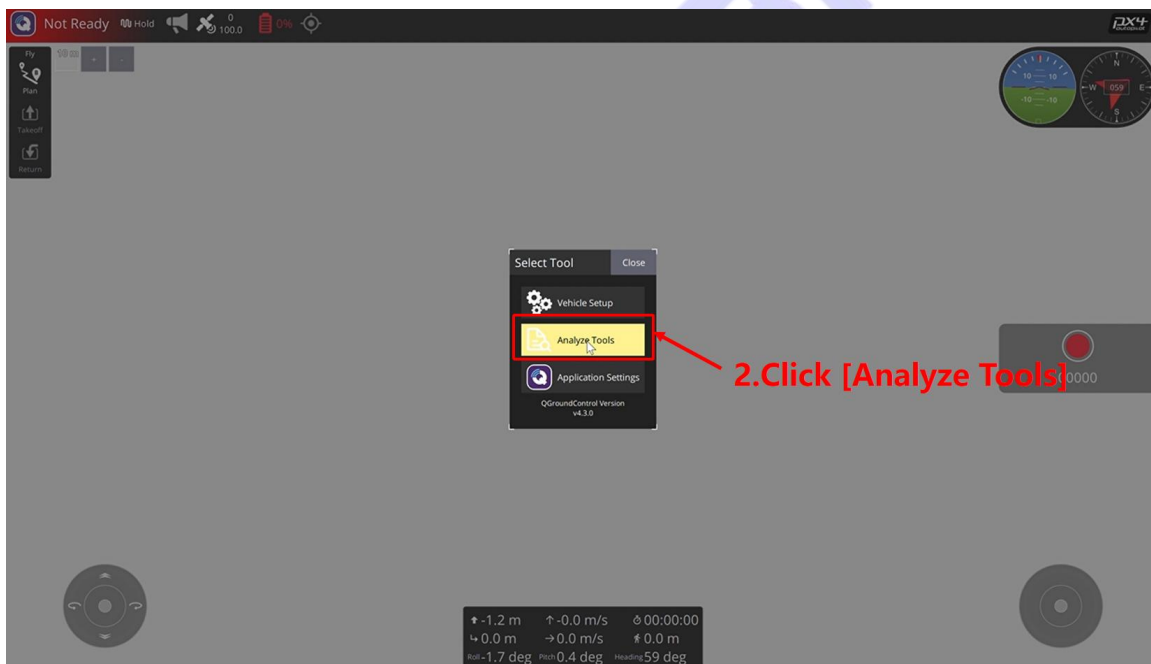
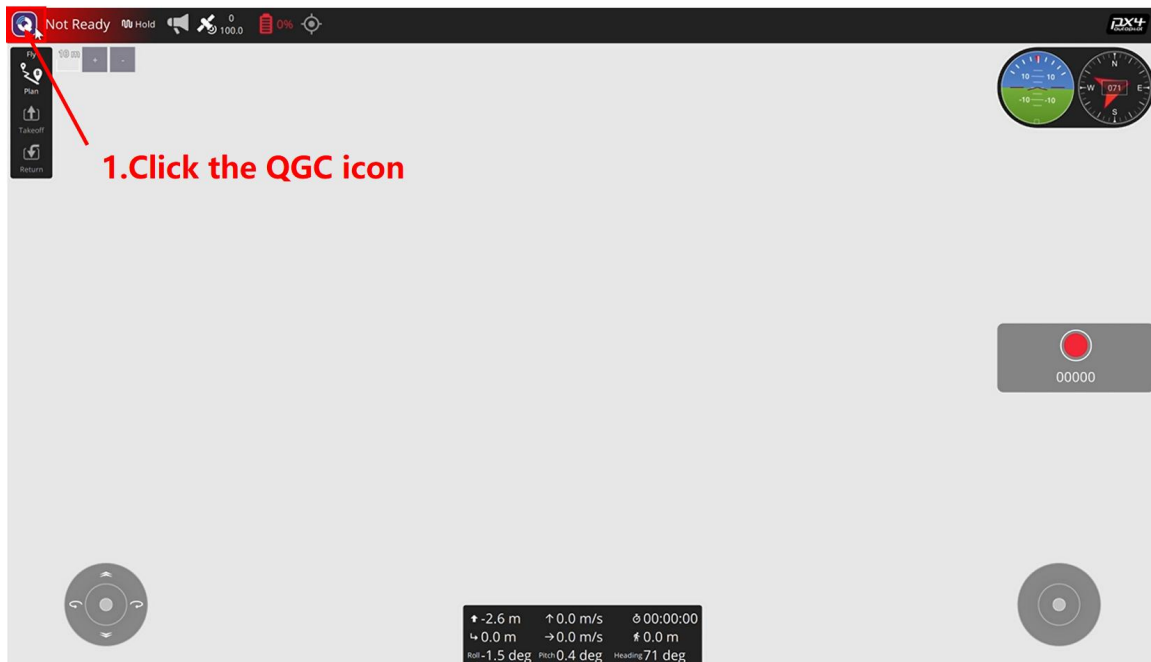
4.3.2 Flight Controller Side Diagnostics

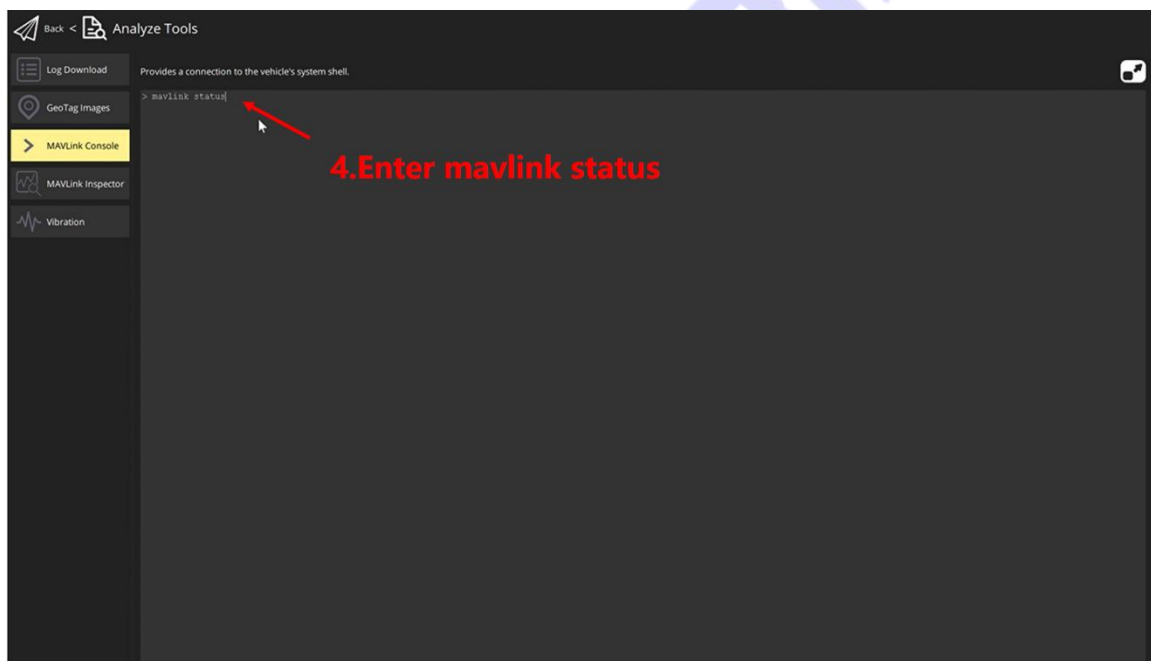
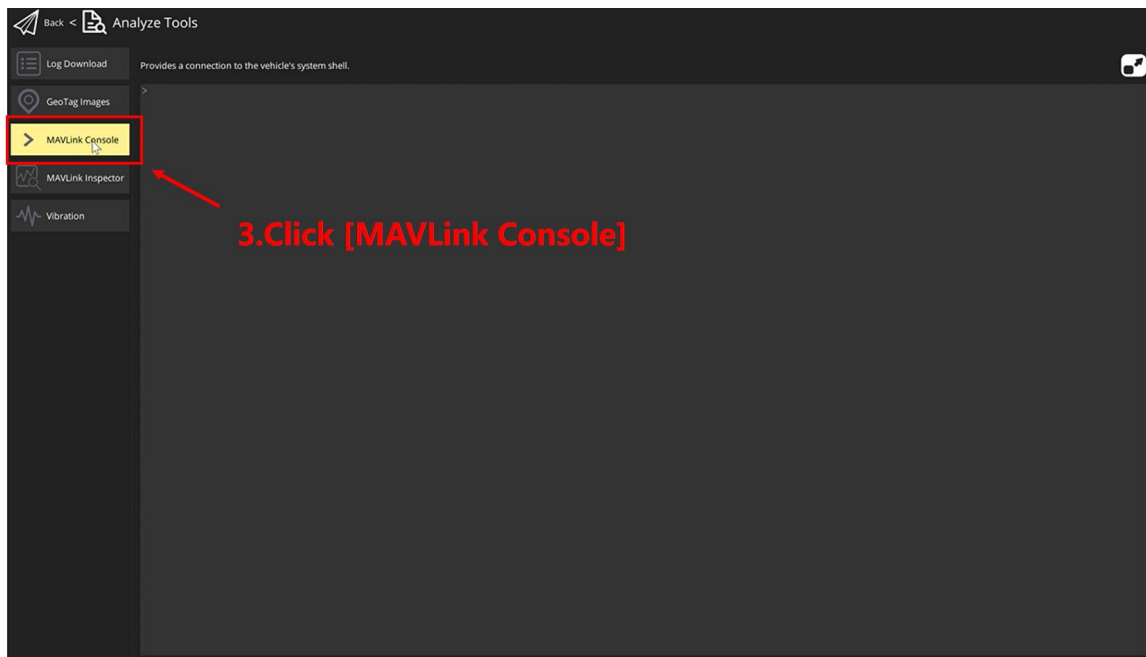
1. Check the MAVLink serial port

In QGC -> Analyze Tools -> MAVLink Console, enter **mavlink status** to check whether the flight controller serial port is set to the correct port, whether the baud rate is correct, and whether messages from the VIO are being received.

If **rx** shows **0.0B/s**, it means no data is being received. Check the serial wiring order and the communication port configuration on the flight controller.

If the baud rate is not **921600**, change the baud rate accordingly and try again.





```

Back < Analyze Tools

Log Download Provides a connection to the vehicle's system shell.
GeoTag Images
MAVLink Console
MAVLink Inspector
Vibration

nsh: sysinit: fopen failed: No such file or directory
Nuttxhell (Nsh) Nuttx-11.0.0
nsh: mavlink status

instance #0:
mavlink chan: #0
type: GENERIC LINK OR RADIO
flow control: ON
rates:
  tx: 55735.5 B/s
  txerr: 0.0 B/s
  tx rate mult: 1.000
  tx rate max: 55735.5 B/s
  rx: 2318.4 B/s
  rx loss: 15.5%
Received Messages:
  sysid: 1, compid: 195, Total: 14380 (lost: 222940)
  FTP enabled: YES, TX enabled: YES
  mode: Onboard
  Forwarding: Off
  MAVLink version: 2
  transport protocol: serial (/dev/ttyS5 921600)
  ping statistics:
    last: 6.20 ms
    mean: 5.73 ms
    max: 8.26 ms
    min: 1.40 ms
    dropped packets: 0

instance #1:
GCS heartbeat valid
mavlink chan: #1
type: GENERIC LINK OR RADIO
flow control: ON
rates:
  tx: 34032.2 B/s
  txerr: 0.0 B/s
  tx rate mult: 1.000
  tx rate max: 921600 B/s
  rx: 935.5 B/s
  rx loss: 83.0%
Received Messages:
  sysid: 145, compid: 190, Total: 14482 (lost: 1202103)
  FTP enabled: YES, TX enabled: YES
  mode: Normal
  Forwarding: On
  MAVLink version: 2
  transport protocol: serial (/dev/ttyS5 921600)
  ping statistics:
    last: 4.51 ms
    mean: 4.15 ms
    max: 5.20 ms
    min: 1.55 ms
    dropped packets: 0

```

2. Check VIO messages on the flight controller side

In QGC -> Analyze Tools -> MAVLink Console, enter **uorb top** to display all internal bus messages, as shown in Figure 4. Check whether the vehicle_visual_odometry message exists and whether the publishing frequency is **30 Hz**.

```

Log Download Provides a connection to the vehicle's system shell.
GeoTag Images
MAVLink Console
MAVLink Inspector
Vibration

nsh: sysinit: fopen failed: No such file or directory
Nuttxhell (Nsh) Nuttx-11.0.0
nsh: mavlink status

instance #1:
GCS heartbeat valid
mavlink chan: #1
type: GENERIC LINK OR RADIO
flow control: ON
rates:
  tx: 34032.2 B/s
  txerr: 0.0 B/s
  tx rate mult: 1.000
  tx rate max: 921600 B/s
  rx: 935.5 B/s
  rx loss: 83.0%
Received Messages:
  sysid: 145, compid: 190, Total: 14482 (lost: 1202103)
  FTP enabled: YES, TX enabled: YES
  mode: Normal
  Forwarding: On
  MAVLink version: 2
  transport protocol: serial (/dev/ttyS5 921600)
  ping statistics:
    last: 4.51 ms
    mean: 4.15 ms
    max: 5.20 ms
    min: 1.55 ms
    dropped packets: 0

instance #2:
GCS heartbeat valid
mavlink chan: #2
type: USB CDC
flow control: OFF
rates:
  tx: 41284.4 B/s
  txerr: 0.0 B/s
  tx rate mult: 1.000
  tx rate max: 100000 B/s
  rx: 620.8 B/s
  rx loss: 0.0%
Received Messages:
  sysid: 255, compid: 190, Total: 9848 (lost: 0)
  FTP enabled: YES, TX enabled: YES
  mode: Onboard
  Forwarding: On
  MAVLink version: 2
  transport protocol: serial (/dev/ttyACM0 92000000)
  ping statistics:
    last: 4.61 ms
    mean: 2.69 ms
    max: 117081.79 ms
    min: 0.24 ms
    dropped packets: 0

nsh> uorb top

```

Log Download	Provides a connection to the vehicle's system shell.			
GeoTag Images	estimator_status	0	13	111 1 136
	estimator_status_flags	0	8	1 1 96
	failsafe_flags	0	5	1 1 88
	failure_detector_status	0	2	2 1 24
MAVLink Console	magnetometer_bias_estimate	0	2	50 1 64
	manual_control_input	0	1	20 1 72
	manual_control_input	1	1	25 1 72
	manual_control_setpoint	0	7	20 1 72
MAVLink Inspector	position_setpoint_triplet	0	6	22 1 248
	px4io_status	0	1	1 1 144
	rate_ctrl_status	0	1	667 1 48
	rtl_status	0	1	1 1 16
Vibration	rtl_time_estimate	0	5	1 1 24
	sensor_accel	0	9	664 8 48
	sensor_accel	1	8	800 8 48
	sensor_accel_raw	0	1	222 1 24
	sensor_baro	0	6	70 4 32
	sensor_combined	0	5	221 1 48
	sensor_gps	0	5	8 1 144
	sensor_gyro	0	10	667 8 48
	sensor_gyro	1	8	800 8 48
	sensor_gyro_fifo	0	2	667 4 224
	sensor_gyro_raw	0	1	667 1 24
	sensor_mag	0	7	46 4 40
	sensor_mag	1	6	46 4 40
	sensor_preflight_mag	0	3	46 1 16
	sensors_status_imu	0	4	221 1 96
	system_power	0	3	100 1 40
	telemetry_status	0	2	1 1 88
	telemetry_status	1	2	1 1 88
	telemetry_status	2	2	1 1 88
	trajectory_setpoint	0	3	50 1 64
	vehicle_acceleration	0	3	221 1 48
	vehicle_air_data	0	13	48 1 48
	vehicle_angular_velocity	0	11	667 1 40
	vehicle_attitude	0	12	221 1 56
	vehicle_attitude_setpoint	0	6	111 1 56
	vehicle_command_ack	0	5	10 4 24
	vehicle_constraints	0	1	50 1 24
	vehicle_control_mode	0	20	2 1 24
	vehicle_gps_position	0	6	221 1 56
	vehicle_imu	0	4	200 1 56
	vehicle_imu	1	5	10 1 136
	vehicle_imu_status	0	23	1 1 24
	vehicle_land_detected	0	37	111 1 184
	vehicle_local_position	0	8	111 1 84
	vehicle_local_position_setpoint	0	7	12 1 40
	vehicle_magnetometer	0	2	111 1 112
	vehicle_odometry	0	5	221 1 40
	vehicle_rates_setpoint	0	48	2 80
	vehicle_status	0	6	667 1 32
	vehicle_thrust_setpoint	0	2	667 1 32
	vehicle_torque_setpoint	0	2	667 1 32
	vehicle_visual_odometry	0	2	31 1 112

2. Check whether the vehicle_visual_odometry message exists and whether the publishing frequency is 30 Hz

In QGC -> Analyze Tools -> MAVLink Console, enter listener vehicle_visual_odometry.

[illegible]

Log Download

Provides a connection to the vehicle's system shell.

GeoTag Images

MAVLink Console

MAVLink Inspector

Vibration

Topic	Rate	Size	QoS	Priority
sensor_gps	0	5	0	1
sensor_gyro	0	10	667	8
sensor_gyro	1	8	800	8
sensor_gyro_fifo	0	2	667	4
sensor_gyro_raw	0	1	667	1
sensor_mag	0	7	46	4
sensor_mag	1	6	46	4
sensor_preflight_mag	0	3	46	1
sensors_status_imu	0	4	221	1
system_power	0	3	100	1
telemetry_status	0	2	1	1
telemetry_status	1	2	2	1
telemetry_status	2	2	2	1
trajectory_setpoint	0	3	50	1
vehicle_acceleration	0	2	221	1
vehicle_air_data	0	15	19	1
vehicle_angular_velocity	0	11	667	1
vehicle_attitude	0	12	221	1
vehicle_attitude_setpoint	0	6	110	1
vehicle_command_ack	0	5	10	4
vehicle_constraints	0	2	50	1
vehicle_control_mode	0	20	3	1
vehicle_gps_position	0	11	8	1
vehicle_imu	0	6	221	1
vehicle_imu	1	4	200	1
vehicle_imu_status	0	9	9	1
vehicle_imu_status	1	5	10	1
vehicle_land_detected	0	2	1	1
vehicle_local_position	0	37	110	1
vehicle_local_position_setpoint	0	8	110	1
vehicle_magnetometer	0	2	110	1
vehicle_odometry	0	2	110	1
vehicle_rates_setpoint	0	48	3	1
vehicle_status	0	6	667	1
vehicle_thrust_setpoint	0	2	110	1
vehicle_torque_setpoint	0	2	30	1
vehicle_visual_odometry	0	2	30	1

```

nsh>
nsh> listener vehicle_visual_odometry

TOPIC: vehicle_visual_odometry
vehicle_visual_odometry
timestamp: 608846080 (0.028228 seconds ago)
timestamp_sample: 608846079 (1 us before timestamp)
position: [0.00101, -0.00114, 0.88697]
q: [1.00000, 0.00000, 0.00000, 0.00000] (Roll: 0.0 deg, Pitch: -0.0 deg, Yaw: 0.0 d
velocity: [nan, nan, nan]
angular_velocity: [nan, nan, nan]
position_variance: [121.00000, 121.00000, 0.00000]
orientation_variance: [0.00000, 0.00000, 0.00000]
velocity_variance: [nan, nan, nan]
pose_frame: 1
velocity_frame: 0
reset_counter: 0
quality: 0
nsh>

```

2. Can check the timestamp, position information, and position variance

4.4 Reboot the Flight Controller

- After the VIO module is connected to the PX4 flight controller, it will automatically configure certain related parameters in PX4 to ensure proper operation of the VIO system and stable flight performance.
- After the VIO module modifies the flight controller parameters, **the flight controller must be rebooted for the changes to take effect.**

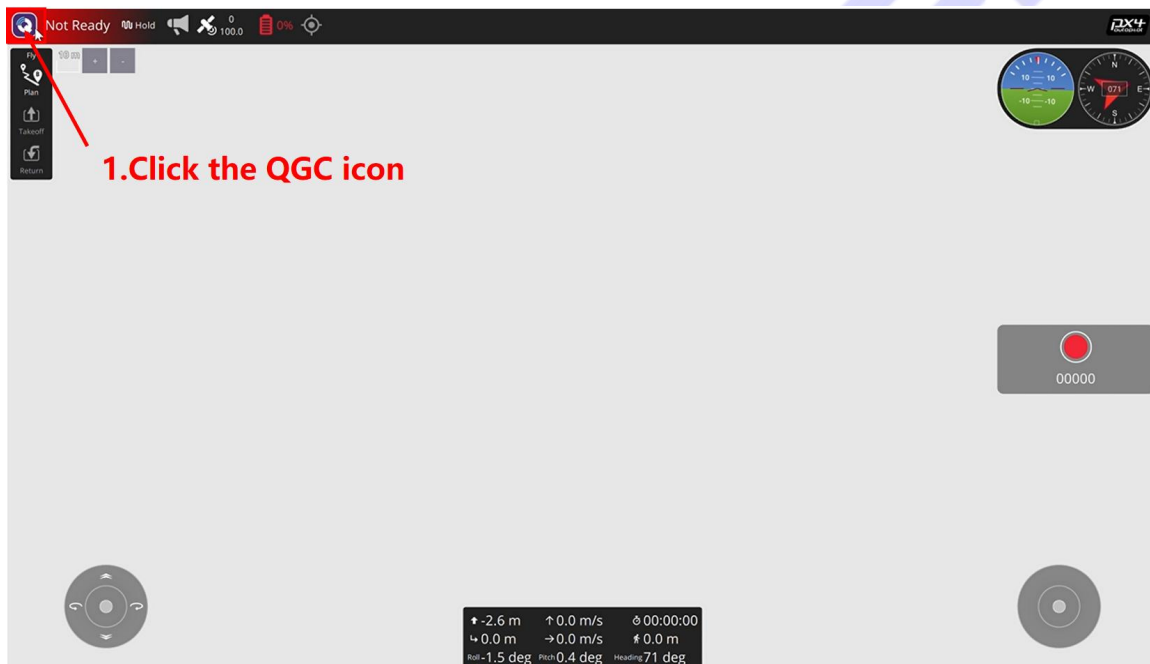
4.5 GPS-denied Flight

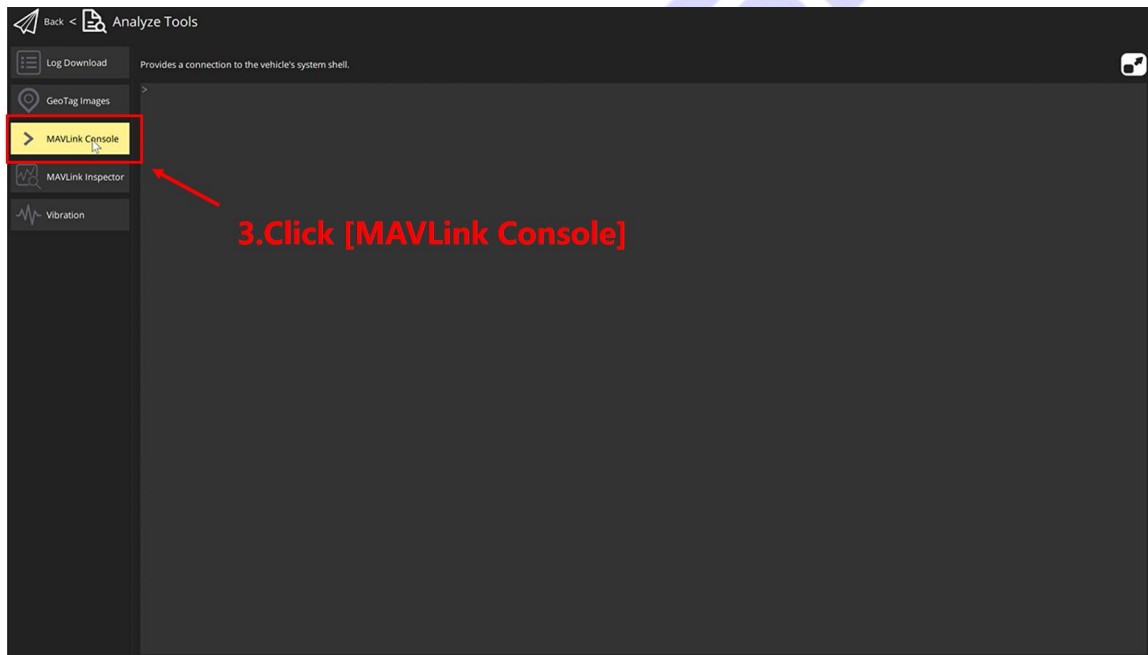
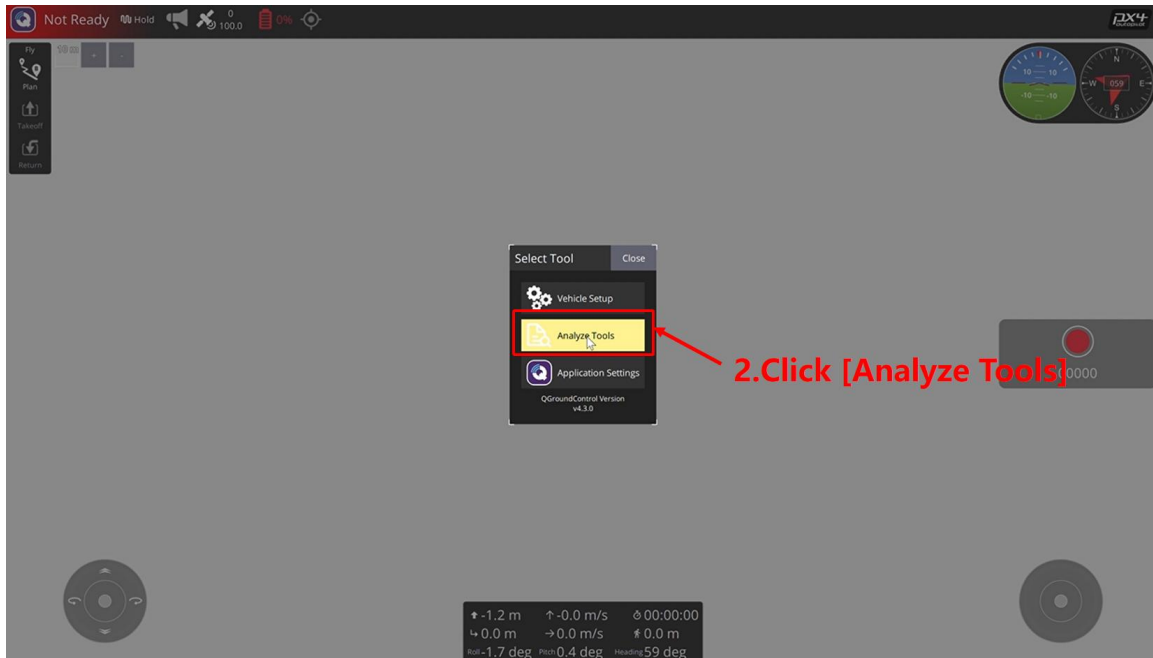
If there is no GPS sensor at all, the drone does not know its current global position. Since this module only provides relative positioning with respect to the takeoff point, the actual latitude, longitude, and altitude of the drone's Home point must be set manually; otherwise, navigation functions cannot be used.

1. In QGC -> Analyze Tools -> MAVLink Console, enter **commander set_ekf_origin** followed by the actual latitude, longitude, and altitude values to be set. For example:

commander set_ekf_origin 12.123123 12.123123 110

2. Enter **listener vehicle_local_position** to verify the settings, as shown in the figure.





Back < Analyze Tools

Log Download Provides a connection to the vehicle's system shell.

GeoTag Images

MAVLink Console

MAVLink Inspector

Vibration

```
nsh: sysinit: fopen failed: No such file or directory
NuttShell (NSH) NuttX-11.0.0
nsh> commander set_ekf_origin 22.7563576 13.426513 110.552
```

lat lon alt

4. Enter the following command to set the 3D coordinates of the Home point:

nsh> commander set_ekf_origin followed by the actual latitude, longitude, and altitude values to be set. For example:

22.7563576 13.426513 110.552

Back < Analyze Tools

Log Download Provides a connection to the vehicle's system shell.

GeoTag Images

MAVLink Console

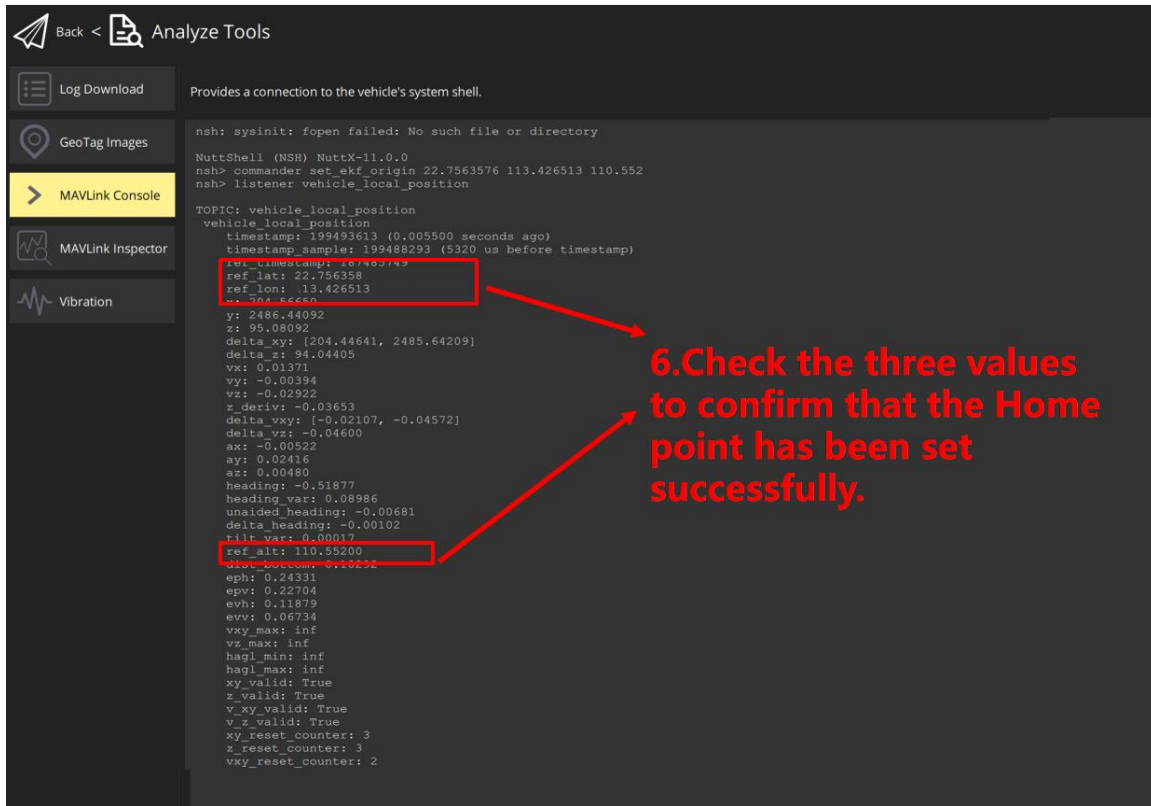
MAVLink Inspector

Vibration

```
nsh: sysinit: fopen failed: No such file or directory
NuttShell (NSH) NuttX-11.0.0
nsh> commander set_ekf_origin 22.7563576 13.426513 110.552
nsh> listener vehicle_local_position
```

5. Enter the following command to check whether the parameters have been written successfully:

nsh> listener vehicle_local_position



4.6 Test Precautions

After completing the above configuration, normal flight testing can be carried out. Please note the following during testing:

- To avoid position drift caused by unstable vision during low-altitude flight, the module only outputs visual positioning data when flying above **3 meters** by default. If you want to disable this limit, enable the **TestOnce** button on the configuration webpage. After enabling it, data will also be output below **3 meters**; however, visual positioning may be unstable at this altitude. It is recommended to use this only for checking data output and not for flight at such low altitudes. After the module is restarted, the restriction of no data output below **3 meters** will be restored.
- The currently supported ground-relative flight altitude range for **N200** is **30 m to 200 m**, and the recommended maximum flight speed is **15 m/s**. When flying too high or too fast, ground feature points move too quickly in the image, which may reduce positioning accuracy.

4.7 Common Troubleshooting

Issue

The flight controller is not receiving messages from the VIO module. No ODOMETRY messages are detected.
EKF not fusing vision

Excessive drift or failure to hold position during outdoor flight above 30 m.

Troubleshooting directions

Baud rate / wiring / module startup

EKF2_EV_CTRL is not configured correctly / abnormal publishing rate / timestamp anomaly.

Camera blur / poor ground texture

5. How an APM Flight Controller Use the VIO Module

This module currently supports **Copter-4.6.0 and above**. Parameter settings may differ in other versions.

To configure the connection parameters between the APM flight controller and the VIO module, you need to log in to the VIO configuration web interface and use **Mission Planner** software.

Before proceeding, ensure that the VIO module is properly powered, the communication serial port is connected to the flight controller, the VIO web interface has been successfully accessed, and the flight controller is connected to Mission Planner.

5.1 Connecting VIO to APM

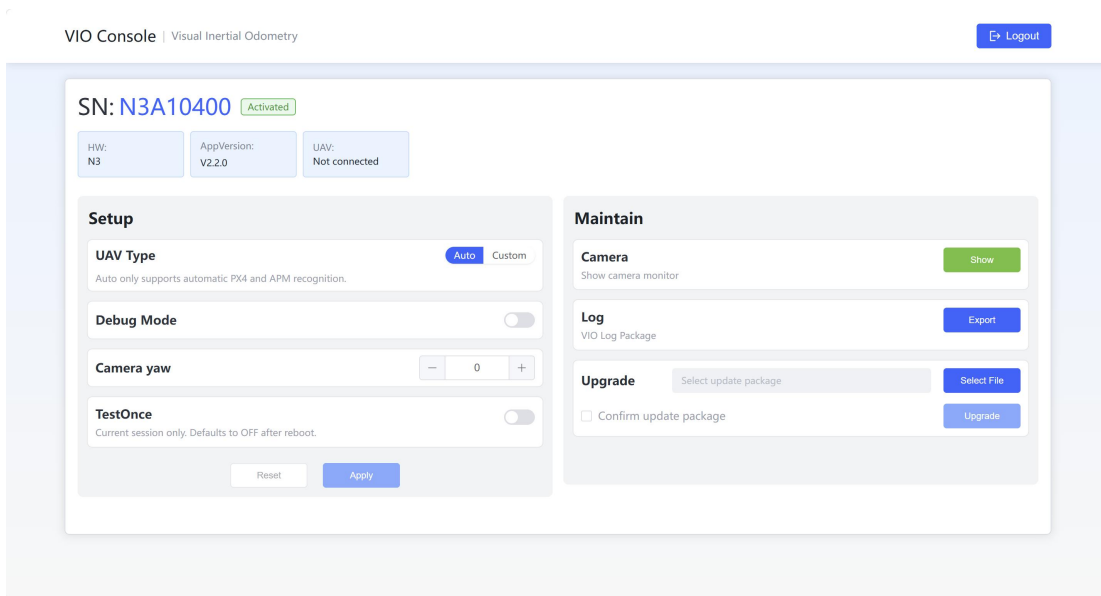
5.1.1 VIO Parameter Configuration

After logging in to the VIO configuration web interface:

Camera Yaw: Enter the actual angle between the VIO and the aircraft's nose. If there is no offset, no change is required.

Other parameters: Keep the default settings unchanged.

If any parameters are modified, click **Apply** to save the settings. Then power off and restart the VIO module. The parameter changes will only take effect after the module reboots.



5.1.2 Flight Controller Communication Parameters

1. Confirm the APM serial port

The physical port labels and internal system identifiers may differ across different APM flight controller models. First, confirm which serial port on the APM flight controller is connected to the VIO module.

Using **Pixhawk 6C Mini** as an example, if the VIO module is connected to the **TELEM 2** port on the APM flight controller, then the serial port to be configured in the following steps should be **SERIAL2**.

UART Mapping

- SERIAL0 -> USB
- SERIAL1 -> UART7 (Telem1) RTS/CTS pins
- **SERIAL2 -> UART5 (Telem2) RTS/CTS pins**
- SERIAL3 -> USART1 (GPS1)
- SERIAL4 -> UART8 (GPS2)
- SERIAL5 -> USART2 (Telem3) RTS/CTS pins (not included on 6C Mini)
- SERIAL6 -> USART3 (USER) (Debug p)
- SERIAL7 -> USB (can be used for SLCAN with protocol change)

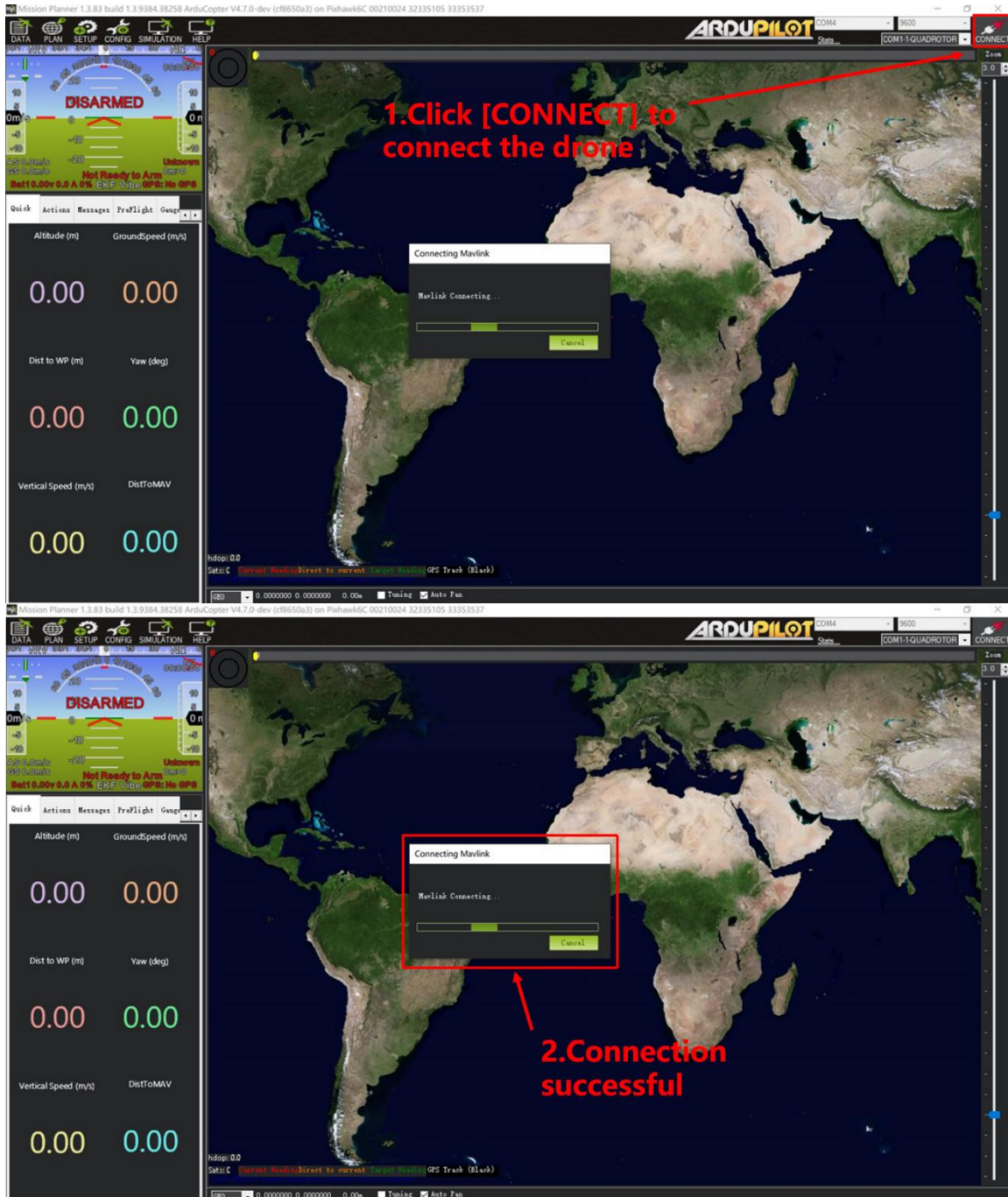
TELEM 2 corresponds to SERIAL2

The official APM documentation link is provided below for users to refer to as needed:

<https://ardupilot.org/plane/docs/common-holybro-pixhawk6C.html>

2. Set the serial port baud rate and communication protocol

- 1) Connect the flight controller to **Mission Planner**, open **Config/Setup**, find **SERIAL2_BAUD**, and set it to **921600**.
- 2) Set **SERIAL2_PROTOCOL** to **MAVLink1**.
- 3) Click **Write Params**, then restart the flight controller and the VIO module.



[illegible]

Mission Planner 1.3.83 build 1.3.9384.38258 ArduCopter V4.7.0-dev (cf8650a3) on Pixhawk6C 00210024 32335105 33353537

ARDUPILOT COM4 9600 COM4-1 QUADROTOR DISCONNECT

GeoFence
Basic Tuning
Extended Tuning
Onboard OSD
MAVtp
User Params
Full Parameter List
Planner
CubeLan 8 port Switch

Name	Δ	Value	Default	Units	Options	Desc	Fav
SERIAL2_BAUD	921	57			921600	The baud rate of the Telem2 port. Most stm32-based boards can support rates of up to 1500. If you setup a rate you cannot support and then can't connect to your board you should load a firmware from a different controller board. There will be a warning if you do this.	☐ Load from file ☐ Save to file ☐ Write Params ☐ Refresh Params ☐ Compare Params
SERIAL2_OPTIONS	0	0			1200 4800 9600 19200 38400 76800 111100 115200 230400 256000 409600 500000 800000	Control over UART options. The InvertTX option controls invert of the receive pin. The InvertTX option controls invert of the transmit pin. The HalfDuplex option controls half duplex (inverted) mode, where both transmit and receive is done on this hardware pin. This option allows the TX and RX pins to be swapped and receive is done on this hardware pin. This option allows the TX and RX pins to be swapped and receive is done on this hardware pin. This option allows the TX and RX pins to be swapped and receive is done on this hardware pin.	☐ All Units are in raw format with no scaling ☐ Load Preset ☐ Reset to Default ☐ Search
SERIAL2_PROTOCOL	1	2			1. 5MBaud 12. 5MBaud	Control what protocol to use on the Telem2 port. Note that the Frsky options require external converter hardware. See the wiki for details.	☐ Modified ☐ None Default

6. Click to modify to 921600

Mission Planner 1.3.83 build 1.3.9384.38258 ArduCopter V4.7.0-dev (cf8650a3) on Pixhawk6C 00210024 32335105 33353537

ARDUPILOT COM4 9600 COM4-1 QUADROTOR DISCONNECT

GeoFence
Basic Tuning
Extended Tuning
Onboard OSD
MAVtp
User Params
Full Parameter List
Planner
CubeLan 8 port Switch

Name	Δ	Value	Default	Units	Options	Desc	Fav
SERIAL2_BAUD	921	57			12000000 1200 57600	The baud rate of the Telem2 port. Most stm32-based boards can support rates of up to 1500. If you setup a rate you cannot support and then can't connect to your board you should load a firmware from a different controller board. There will be a warning if you do this.	☐ Load from file ☐ Save to file ☐ Write Params ☐ Refresh Params ☐ Compare Params
SERIAL2_OPTIONS	0	0			1200 4800 9600 19200 38400 76800 111100 115200 230400 256000 409600 500000 800000	Control over UART options. The InvertTX option controls invert of the receive pin. The InvertTX option controls invert of the transmit pin. The HalfDuplex option controls half duplex (inverted) mode, where both transmit and receive is done on this hardware pin. This option allows the TX and RX pins to be swapped and receive is done on this hardware pin. This option allows the TX and RX pins to be swapped and receive is done on this hardware pin.	☐ All Units are in raw format with no scaling ☐ Load Preset ☐ Reset to Default ☐ Search
SERIAL2_PROTOCOL	1	2			MAVLink1	Control what protocol to use on the Telem2 port. Note that the Frsky options require external converter hardware. See the wiki for details.	☐ Modified ☐ None Default

7. Find SERIAL2_PROTOCOL

8. Set to MAVLink1

Mission Planner 1.3.83 build 1.3.9384.38258 ArduCopter V4.7.0-dev (cf8650a3) on Pixhawk6C 00210024 32335105 33353537

ARDUPILOT COM4 9600 COM4-1QUADROTOR DISCONNECT

GeoFence
Basic Tuning
Extended Tuning
Onboard OSD
MAVLink
User Params
Full Parameter List
Planner
Cubelink B port Switch

Name	Δ	Value	Default	Units	Options	Desc	Save
SERIAL2_BAUD	521	57			1:20000000 1:1200 5:2400	The baud rate of the Telem2 port. Most stm32-based boards can support rates of up to 1500. If you setup a rate you cannot support and then can't connect to your board you should load a firmware from a different controller series. This will reset all serial connections to default.	Save to File
SERIAL2_OPTIONS	0	0				Control over UART options. The InvertTX option controls invert of the receive pin. The InvertTX option controls invert of the transmit pin. The HalfDuplex option controls half duplex (one-way) mode, where both transmit and receive happen on the same pin. This option controls the TX and RX pins to both transmit and receive on the same pin. This option controls the TX and RX pins to both transmit and receive on the same pin. This option controls the TX and RX pins to both transmit and receive on the same pin.	Write Params
SERIAL2_PROTOCOL	1	2			MAVLink1	Control what protocol to use on the Telem2 port. Note that the Frsky options require external converter hardware. See the wiki for details.	Refresh Params

9.Click [Write Params]

10.Restart the flight controller and the VIO module

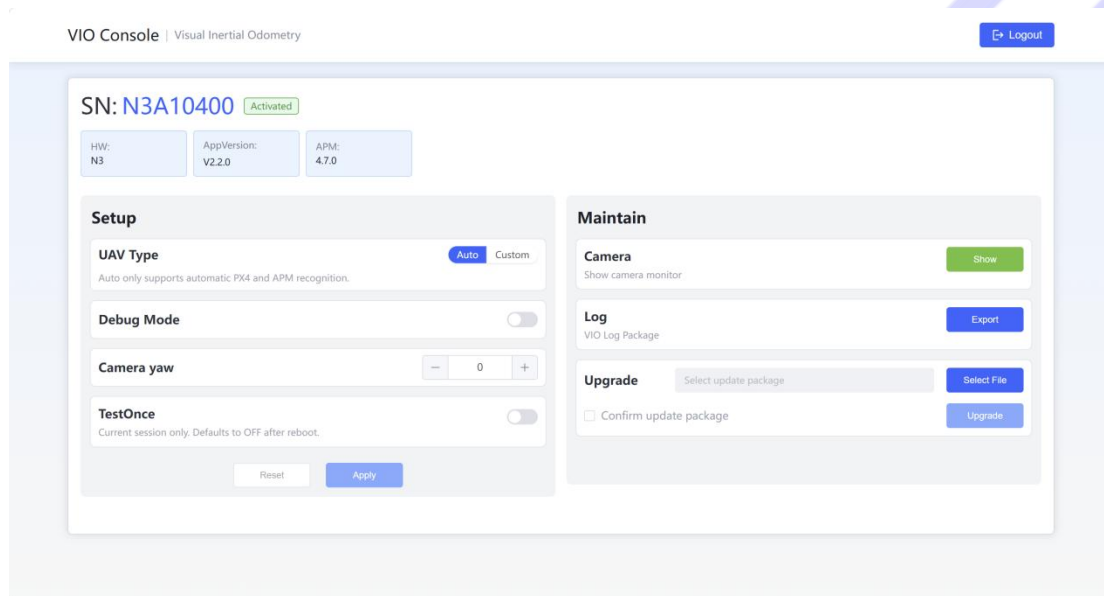
Load from File
Save to File
Write Params
Refresh Params
Configure Params
All Units are in raw format with no scaling
SDR_Test.param
Load Preset
Reset to Default
Search
Modified
None Default

5.1.3 Connection Status Verification

(1) Check on the VIO side

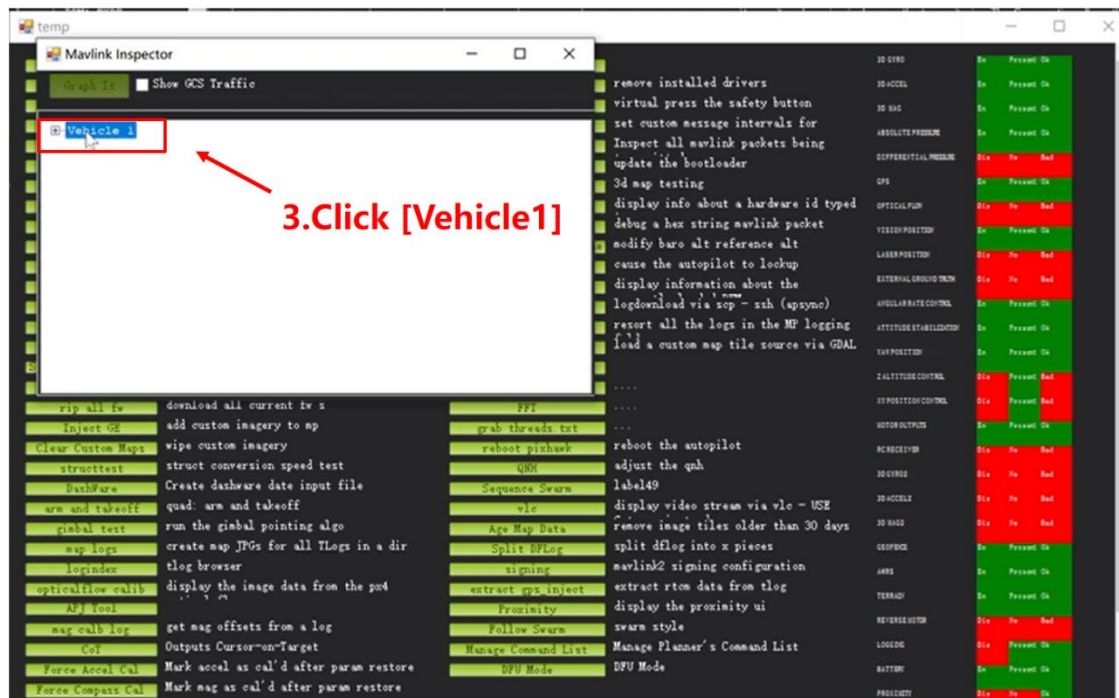
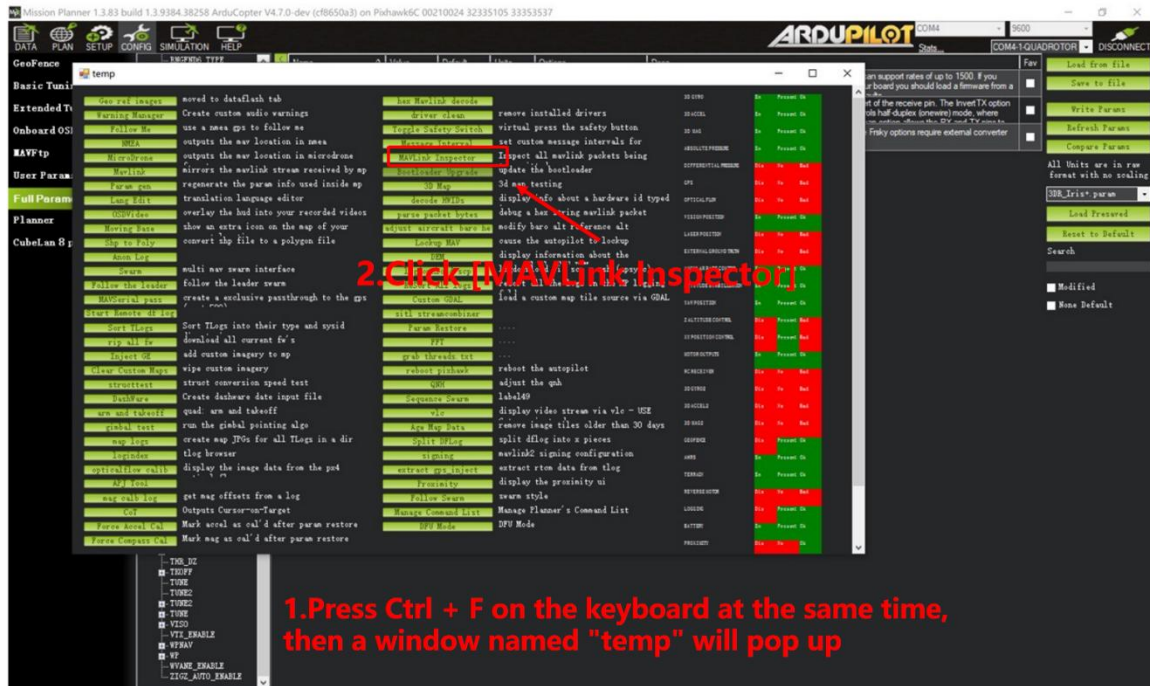
Log in to the VIO configuration web interface and check whether the VIO has successfully retrieved the flight controller version information. If the version information is displayed, it indicates that the VIO and the flight controller are connected successfully and communication is functioning normally.

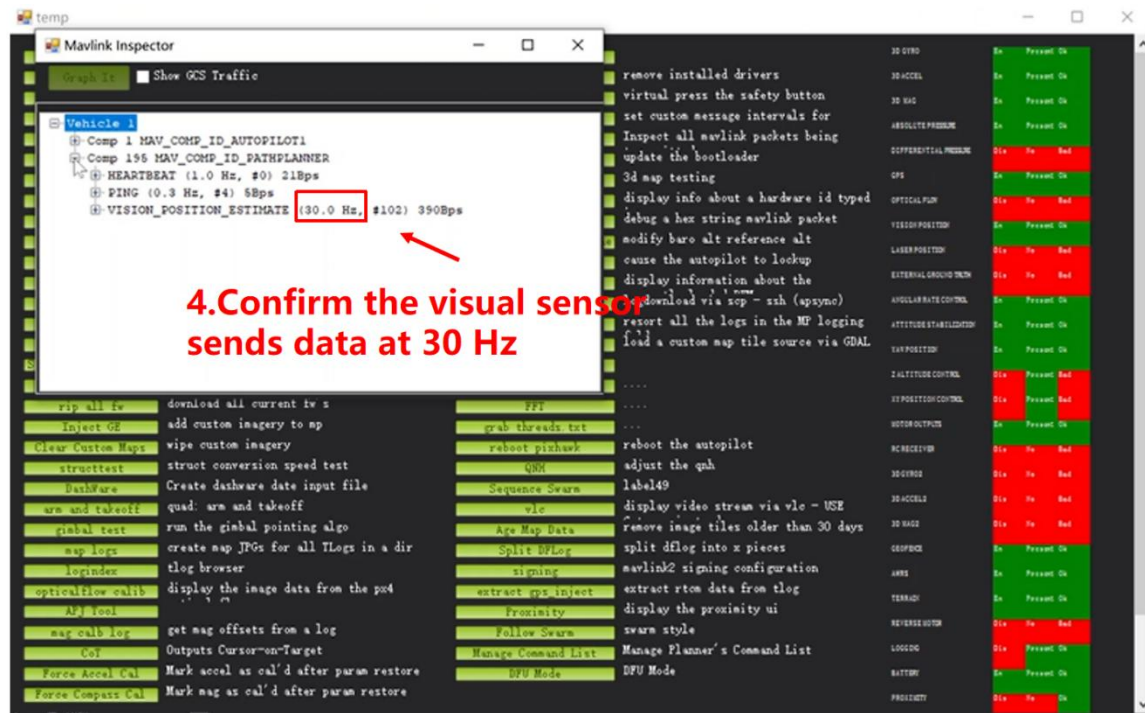
If no flight controller information is detected, check the serial wiring order, baud rate settings, and whether the standard MAVLink protocol is enabled on the flight controller side.



(2) Check on the flight controller side

On Mission Planner, press **Ctrl + F** on the keyboard, and a window named **"temp"** will pop up. Find and click **MAVLink Inspector**, then check whether the data sent by the visual sensor is at **30 Hz**.





5.2 APM VIO Test Procedures

On APM, GPS and VIO positioning cannot be fused and only one can be used at a time. Therefore, users can choose to disable GPS and rely entirely on VIO for positioning, or manually switch between GPS and VIO. These two configuration methods are different.

For initial use, it is recommended to use **GPS + VIO switching** to ensure safety.

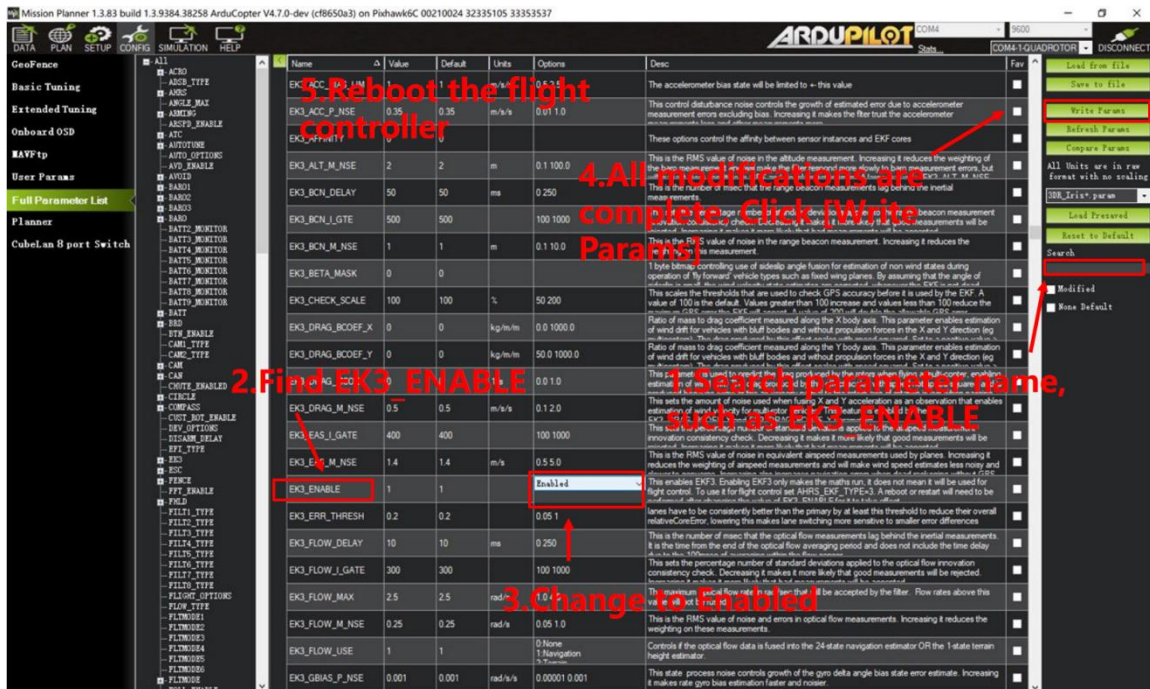
5.2.1 GPS + VIO Manual Switching Test

(1) APM Parameter Configuration

- Enable **EKF3** and disable **EKF2**. In newer APM versions, EKF2 may not be available, in which case no configuration is required.
- Keep **EK3_SRC1** set to GPS-related sources, and configure **EK3_SRC2** for VIO. Since VIO only outputs **POSXY**, set **POSZ** to the barometer and **YAW** to the compass. No velocity-related outputs are provided.
- Configure a three-position switch for positioning source selection, for example **RC8**. When the switch is in the low position, **EK3_SRC1** is used; in the middle position, **EK3_SRC2** is used; and in the high position, **EK3_SRC3** is used. In this example, **SRC3** is not configured, so do not switch to the high position.
- The parameters mentioned above are summarized in the table below. Please modify the corresponding parameter values accordingly.

Parameters need to be modified	value	Option
EK3_ENABLE	1	Enabled
EK2_ENABLE	0	Disabled (Ignore if not present)
AHRS_EKF_TYPE	3	\ (Ignore if not present)
EK3_SRC2_POSXY	6	ExternalNav
EK3_SRC2_POSZ	1	Baro
EK3_SRC2_VELXY	0	None
EK3_SRC2_VELZ	0	None
EK3_SRC2_YAW	1	Compass
EK3_SRC_OPTIONS	0	Disable the fusing of all velocities
RC8_OPTION	90	EKF Pos Source / EKF Source Set

- An example of the modification steps is shown in the figure below. After all parameter values in the table have been updated, click **[write params]**. Once the write is complete, **restart the flight controller**.



(2) Recommended Test Procedure for GPS + VIO

When both GPS and VIO navigation are configured on the drone, note that VIO performs poorly at low altitudes. Therefore, it is recommended to use GPS for takeoff and climb to a certain height before switching to VIO to evaluate its positioning performance.

- Keep **RC8** in the low position. Power on the drone, wait for GPS satellite convergence, and prepare for takeoff.
- Manually take off vertically to an altitude above 10 meters, ensuring stable position hold.
- Switch **RC8** to the middle position to use VIO for positioning and maintain hover.
- Use the control sticks to maneuver the drone forward, backward, left, and right, and observe positioning performance.
- Set a mission/waypoints and observe the positioning accuracy during the flight.

5.2.2 VIO-only (No GPS) Test

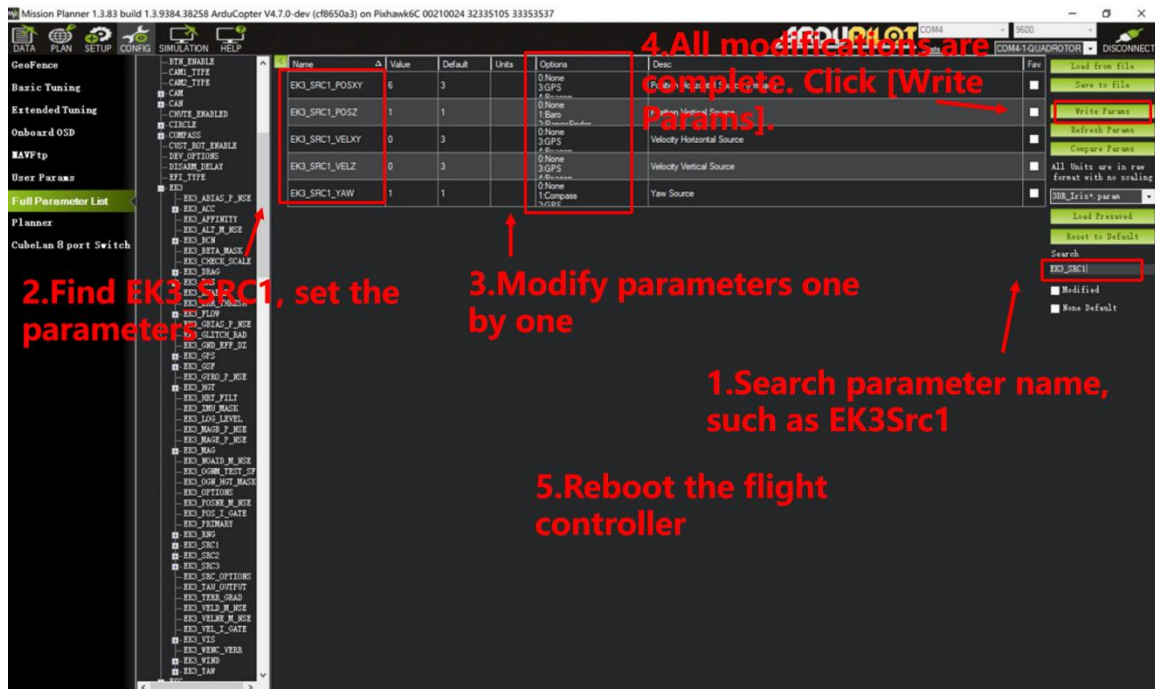
(1) APM Parameter Configuration

- Enable **EKF3** and disable **EKF2**. In newer APM versions, **EKF2** may not be available, in which case no configuration is required.
- When VIO is the only positioning source, set **EK3_SRC1_** to VIO. Since VIO only outputs **POSXY**, configure **POSZ** to use the barometer and **YAW** to use the compass. No velocity-related outputs are provided.
- Disable **GPS1** to avoid pre-arm checks such as “EKF3 waiting for GPS config data”. If **GPS2** is configured, disable it as well.

- The parameters mentioned above are summarized in the table below. Please modify the corresponding values accordingly.

Parameters need to be modified	value	Option
EK3_ENABLE	1	Enabled
EK2_ENABLE	0	Disabled (Ignore if not present)
AHRS_EKF_TYPE	3	\
EK3_SRC1_POSXY	6	ExternalNav
EK3_SRC1_POSZ	1	Baro
EK3_SRC1_VELX Y	0	None
EK3_SRC1_VELZ	0	None
EK3_SRC1_YAW	1	Compass
GPS1_TYPE	0	None (disable GPS 1)

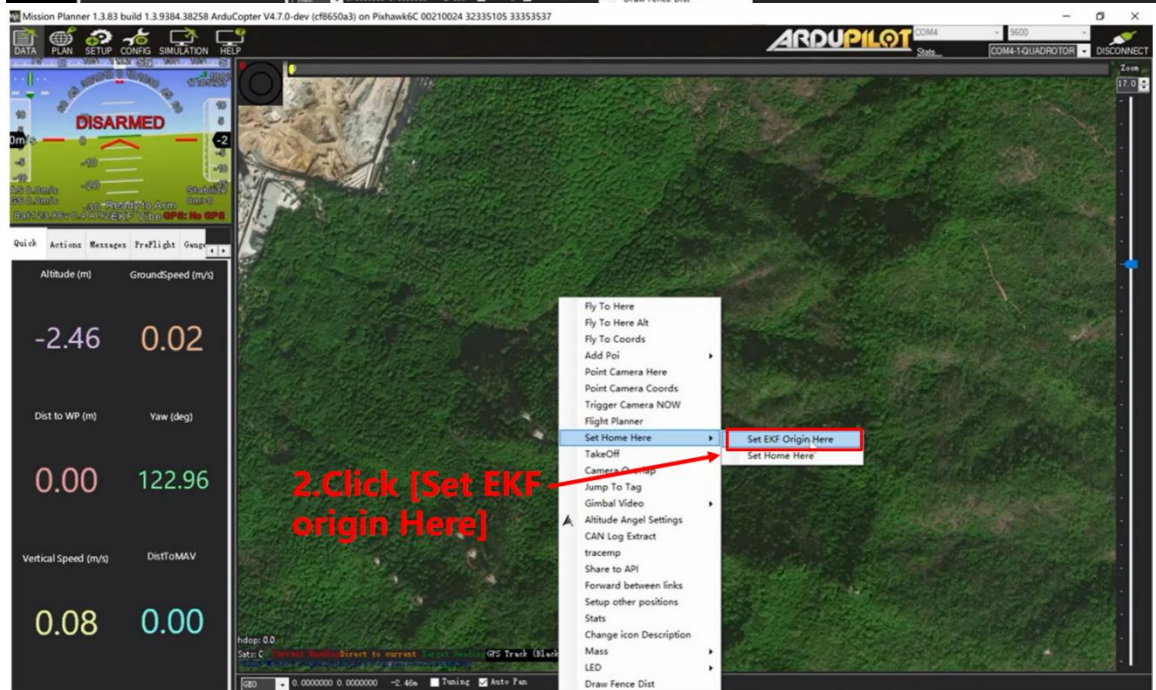
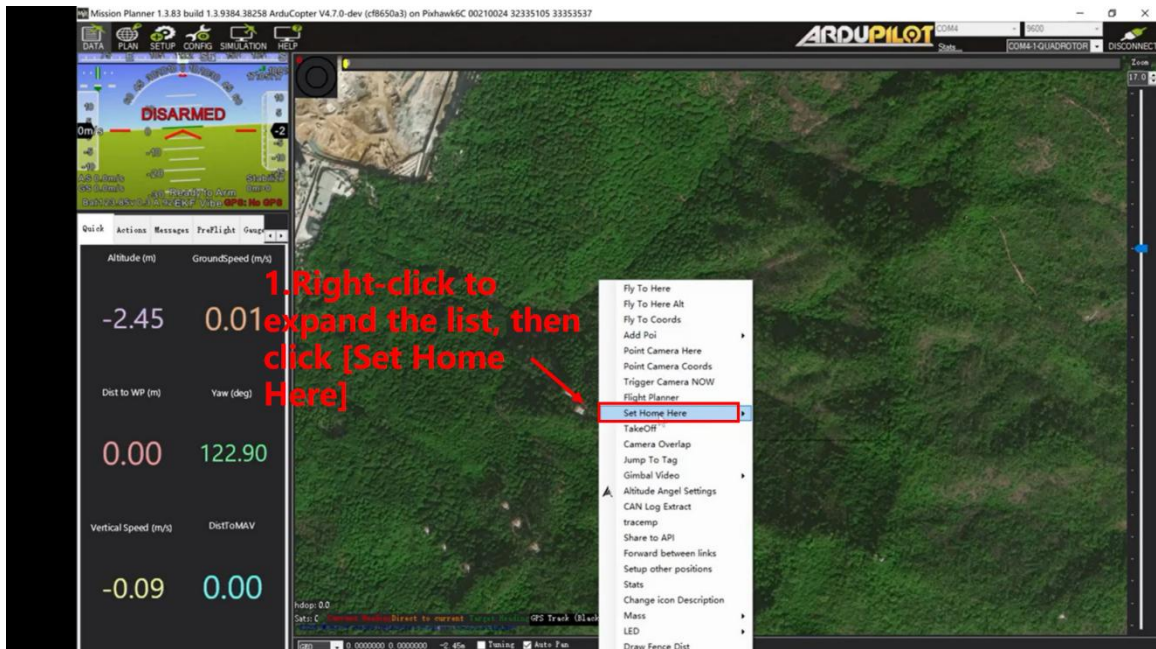
- An example of the modification steps is shown in the figure below. After all parameter values in the table have been updated, click **[write params]**. Once the write is complete, **restart the flight controller**.

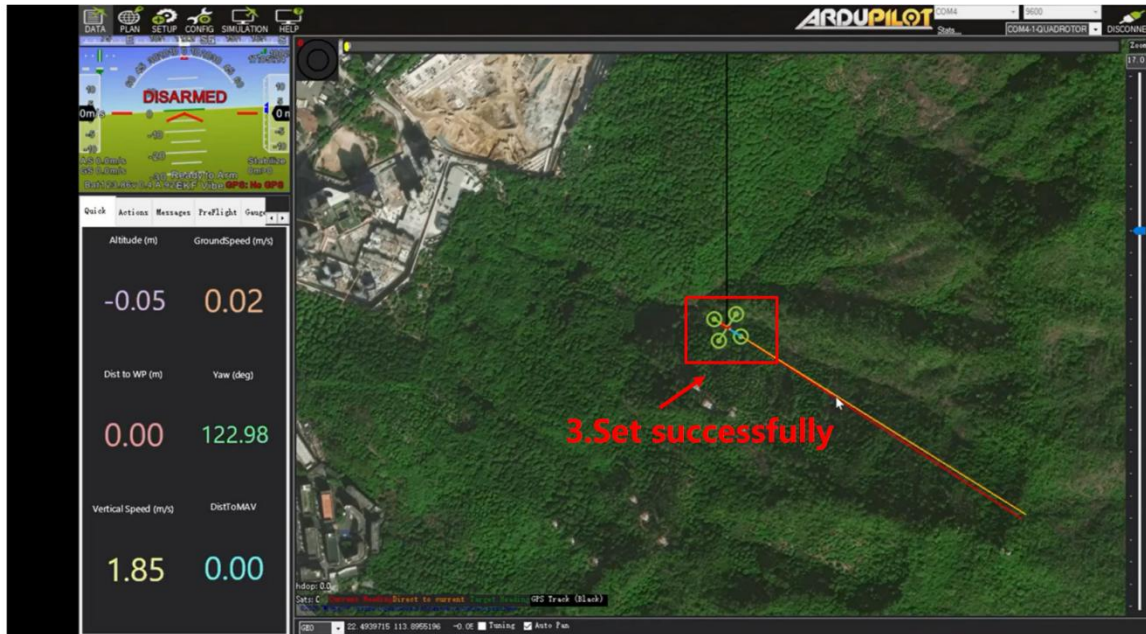


(2) World Reference Point

World Reference Point: Without a world reference point, external positioning data cannot be properly fused by the **EKF**. Normally, this reference point is provided after GPS stabilizes. Since GPS is not used in this case, it must be set manually.

- In Mission Planner, locate the drone's current position on the map, right-click, and select **Set Home Here** → **Set EKF Origin Here**.
- The **EKF Origin** is used to establish the coordinate system, while **Home** is the return point used during RTL. In most cases, these two positions should be the same.





- The reference point coordinates can also be set via commands in MAVProxy. Further research is left to the user.
- The drone can now perform normal flight, return-to-home, and autonomous mission flights. Since positioning accuracy is lower near the ground, it is recommended to take off vertically to **above 10 meters** before transitioning to horizontal flight.

5.3 Test Notes

After completing the above configuration, normal flight testing can be performed. Please note the following during testing:

- To prevent position drift caused by unstable vision during low-altitude flight, the module only outputs visual positioning data when flying above **3 meters** by default. If this restriction needs to be removed, enable the **TestOnce** button on the configuration webpage. After enabling it, data will also be output below **3 meters**; however, visual positioning will be unstable. It is recommended to use this mode only for data observation and not for flight at such low altitudes. After the module is restarted, the default restriction (no output below **3 meters**) will be restored.
- The N200 supports a ground-relative flight altitude range of 30 m to 200 m, with a recommended maximum flight speed of 15 m/s. When flying too high or too fast, ground feature points move rapidly in the image, which reduces positioning accuracy.

5.4 Common Flight Issues

Phenomenon:

VIO positioning is accurate near the ground, but when the drone climbs to around 40–50 meters, it begins to oscillate or drift significantly. This issue becomes more pronounced in strong wind conditions.

Solution:

This occurs because strong winds cause more aggressive attitude changes, and the relatively high noise from VIO data leads to incorrect state estimation in the flight controller.

To resolve this, reduce the parameter **VISO_POS_M_NSE** from **10 to 5**, thereby increasing the weight of VIO measurements and stabilizing the drone.